

Red Sand

Propellant Lead • Arcadia Rise

15 SOLS • 56 STOPS • GRADE 12 • REDSAND

THE OPENING CARD

The rocket on the pad leaves when Mars and Earth line up, and it has to lift off full. You are the propellant lead at Arcadia Rise, and the propellant is made here out of Martian air and ground. You hold the desk on fifteen critical sols between the first shortfall and flight release. By the last one, the tanks must hold what six people need to leave Mars. Miss the window and they wait twenty-six months.

The tank misses the window

BEFORE THE SOL — TRIAL

Walk the plant while your suit is fresh

Nine modules buried to the eaves along one track, and an ascent vehicle four hundred metres past the last of them. Walk the whole line once early, on a clear sol, because the next time you do it will be in a dust storm on a suit that has been out for three hours already.

PLAN CARD 67W

Sol 291. The ascent vehicle holds 3.9 tonnes of methane and needs 6.6 before the Earth-return window. At the present average the projection lands short. Sundqvist wants more temperature because yesterday's plant was slow; Herrera says a hotter reactor can make methane faster and still leave less methane at equilibrium. Today you walk the whole chain—reaction, catalyst and cold end—and decide which limit is actually costing kilograms.

BRIEFING 19W

The tank reconciliation says the ascent vehicle will miss its methane target unless the plant's real bottleneck is found.

WHAT YOU DECIDE 14W

Establish what the plant is actually limited by before anybody changes a set point.

1.1 What the loop settles at

REACTOR HALL · A MODULE · PROTOCOL

SCENE 45W

The loading projection is still 2,700 kilograms short. Tomás Herrera, the reactor and yield lead, will not accept 'run it hotter' as a plan. Four changes to the loop are on the reactor board, and he wants each effect agreed before anyone touches a valve.

GUIDE 71W

Herrera's point is that these four are not four different rules. They are one rule — Le Châtelier's principle — landing on four different quantities. So work out what each change actually touches. The number of gas molecules on each side? The heat, which is one of the products here? The partial pressure of something taking part, meaning that gas's own share of the total? Or nothing the reaction cares about?

ASK 22W

Match each change to the loop with what it does to the methane at equilibrium — the amount the mixture settles at.

BEHIND THE BUTTON 192W

What Le Châtelier's principle says. An equilibrium is a mixture that has stopped changing overall, because the forward reaction and the back reaction are running at the same speed. Disturb it and it shifts in whichever direction undoes part of the disturbance. That is one rule, and all four of these changes are that rule applied to a different quantity.

Why pressure and temperature are not the same kind of change. Squeezing the loop does not touch the equilibrium constant K — the fixed ratio the mixture settles at for a given temperature. It changes the partial pressures, and the mixture re-settles to that same K with more of the smaller side. Temperature changes K itself. Heat is the only one of these four that moves the ceiling rather than the position along it.

Why an inert gas does nothing. Argon takes no part in the reaction, so adding it raises the total pressure and leaves every partial pressure exactly where it was.

Nothing in the ratio the mixture settles at has moved, so nothing shifts. It is a real change on the gauge and no change at all to the chemistry.

TAKES AS READ

a reaction at equilibrium is still running in both directions · pressure and temperature are the two settings on a gas-phase reactor

VERDICT 89W

Every one of these is the same rule — Le Châtelier's principle: push an equilibrium and it shifts to undo the push. Pressure counts gas molecules, and five go in where three come out, so squeezing the loop favours the smaller side. Heat is one of the products here, so heating the bed lowers the ceiling itself. Condensing the water takes a product away and leaves the mixture short. Argon takes no part, so it raises the total pressure and nobody's partial pressure, and the mixture never notices it.

TAKEAWAY 19W

Four changes, four different effects on an equilibrium. One of the four does not act on it at all.

1.2 Rate is not yield

CATALYST BAY • A PERSON • CHOICE

SCENE 40W

Sundqvist wants the bed at 320 °C tonight, up from 280. At 280 the gas already comes out at its balance point — it has had all the time it needs. Ravneet Kaur, the catalysis engineer, has both charts open.

GUIDE 73W

Here is the thing worth separating before you choose. Heating a bed changes two quantities. One is its rate — how fast the gas reacts. The other is its yield, meaning how much methane you are left with once the mixture stops changing. Two different rules set them, and they are free to move in opposite directions. So read each option and ask whether it keeps them apart or rolls them into one.

ASK 32W

Making methane here is exothermic — it gives off heat. The bed is run 40 degrees hotter, at the same pressure and flow. What happens to the rate, and to the yield?

BEHIND THE BUTTON 219W

What heat does to the rate. A reaction only happens when two molecules hit hard enough, and the size of the push they need is the activation energy. At any temperature some hits clear it and most do not. Heating the bed raises the share that clear it, so more hits work each second. This is true of every reaction — the ones that give off heat and the ones that soak it up alike.

What heat does to the yield. A reaction that gives off heat is called exothermic, and it can be read as making heat right alongside the methane. So heat added from outside acts like extra product. The balance point — where the mixture stops changing — shifts back toward the gases it started from. The yield falls. An endothermic reaction, one that soaks heat up, does the opposite. That is why this direction has to be checked and never assumed.

Why keep rate and yield apart. The rate says how long the plant waits. The yield says what it gets for waiting. A bed that is short of time is fixed by heat. A bed already at its ceiling is only made worse by it. So the reading that settles this decision is whether the gas is coming out at its balance point yet.

TAKES AS READ

heating a reaction makes it go faster • an exothermic reaction gives out heat as it proceeds

VERDICT 89W

Heat does two jobs. Molecules bump into each other harder and more often, so more of those bumps have enough push to react. That is the rate going up, and it happens for every reaction. But this one is exothermic. It gives off heat as it goes. Adding heat from outside is like adding more of what it is already trying to get rid of. So it backs off, and ends up with less methane. That is the yield going down. A hotter bed reaches a lower yield faster.

TAKEAWAY 27W

Temperature acts on how fast a reaction goes and on how much it ends up with. For a reaction that releases heat, those two move opposite ways.

1.3 What freezes before methane condenses

COLD END · A MODULE · CHOICE

SCENE 34W

The cold line takes the product stream from 10 degrees down to 115 kelvin in one run of pipe. Mei-Ling Cho, the cryogenics engineer, will not accept a stream that has skipped the drier.

GUIDE 71W

Cho has a reason for refusing that stream, and it is not fussiness. Follow the cold line down with her. It starts near 10 degrees and finishes at 115 kelvin. The water, the carbon dioxide and the methane in the stream do not all stay gases that far. Each has its own temperature for leaving the gas. Keep that walk down the cold line in mind while you read the four.

ASK 18W

Why does the stream pass a drier and a carbon dioxide trap before it reaches the cold line?

BEHIND THE BUTTON 115W

Why the temperatures need conditions attached. A phase boundary belongs to a pressure as well as a substance. Methane's normal boiling point is about 112 K near one atmosphere; at a different pressure its condensation temperature moves.

Why water and carbon dioxide are removed first. Their dew or frost conditions in this process occur well above the final methane condenser. If either reaches colder metal, it can accumulate as solid where the line was never meant to hold it.

Why a blockage can look like a valve fault. A solid layer growing inside a pipe gradually cuts flow while the refrigerator still holds temperature. From outside, pressure and flow have to rule out the alternatives.

TAKES AS READ

a substance leaves the gas phase at conditions that depend on both temperature and pressure · the final purified methane stream is brought near 1 bar before the condenser, where methane condenses near 112 K · intermolecular forces, and why they set a boiling point — taken as read

VERDICT 92W

The stream has to be cleaned before deep cooling because phase changes depend on the local pressure and composition. Water that escapes the drier can condense or freeze on warmer hardware long before the methane condenser. Residual carbon dioxide can frost out in the dedicated trap before the stream reaches deep cryogenic temperatures. Only after those species are removed is the methane-rich stream brought to the condenser's defined pressure, near 1 bar here, where methane condenses near 112 K. The useful lesson is the order, not a universal list of magic temperatures.

TAKEAWAY 24W

A cold line is a path through pressure and temperature; remove each contaminant before the conditions at which it can foul the next stage.

END OF THE SOL 15W

A plant has a speed and a ceiling, and they are not the same limit.

The hot run

PLAN CARD 65W

Sol 292. The hot run made 11.4 kilograms of methane, up from 9.7. That looks like Sundqvist winning the argument. But the same array must also make hydrogen, dig water and keep the station alive. Today you find how long gas actually sees the catalyst, what charge the electrolysis cells turn into, and whether the extra methane was bought by moving the bottleneck somewhere else.

BRIEFING 23W

One hotter night raised output, but the gain is meaningless until the plant knows whether it came from rate, feed, or borrowed power.

WHAT YOU DECIDE 19W

Find what else the same power could be doing, now that the bed is where it should have been.

2.1 How long a molecule stays in the bed

CATALYST BAY · A MODULE · BALLPARK

SCENE 35W

Kaur wants the contact time worked out at the new flow before anything else is changed. The bed holds 340 litres of packed catalyst and the loop is passing 85 litres a minute through it.

GUIDE 72W

Kaur is asking for a contact time. Contact time is the same thing as residence time: how long a molecule stays inside the bed. A volume and a flow through the bed will give you that. Five numbers are on the board and they are not all about the bed's size. Some describe the conditions in the loop instead. Ask of each whether a length of time could possibly depend on it.

ASK 15W

Estimate the contact time — how long a molecule of gas stays inside the bed.

BEHIND THE BUTTON 215W

What residence time is, and what it is not. It is the time a molecule spends inside the packed bed: its volume divided by the flow through the bed. It says nothing about how quickly the reaction itself runs. A slow reaction given a long contact and a fast one given a short contact can convert the same amount. Conversion is the two of them together.

Why the flow has to be quoted at the reactor's own conditions. A gas volume is not a fixed amount of gas. Squeeze it and the same molecules take up less room; heat it and they take up more. So 85 litres a minute at 12 bar and 593 K is one number of molecules. The same 85 litres a minute out on the surface is quite another. Both numbers here are already quoted inside the bed. That is why the pressure and the temperature are not wanted a second time.

Why the tiles carry labels. A bare number cannot be checked against the relationship it is going into. Reading the label catches a unit that does not match. It also catches a quantity that belongs to a different part of the problem. Do it before you place the tile. That habit is the whole point of this format.

TAKES AS READ

a flow quoted in litres a minute is a volume passing each minute · gas volumes depend on the pressure and temperature they are quoted at · rate constants, and how steeply they climb with temperature — taken as read · stoichiometry as a feed rate: turning a production target into kilograms in — taken as read

VERDICT 82W

Residence time is the contact the reaction gets. Conversion needs two things rather than one. The rate constant says how quickly it reacts at these conditions. Residence time says how long the gas is there to do it in. Push more gas through the same bed and the contact gets shorter. So the plant can raise its flow and lower its conversion

in the same move. Rate and residence time have to be judged together before a higher production setting is accepted.

TAKEAWAY 18W

Conversion depends on how fast the reaction runs and on how long it is given to run for.

2.2 Which electrode does which job

ELECTROLYSIS HALL • A PERSON • PROTOCOL

SCENE 33W

The hall runs two kinds of cell: one splits water, the other splits carbon dioxide at 800 degrees. Rosalind Achebe, the power and electrolysis engineer, has the four parts listed on the board.

GUIDE 78W

Achebe's board is really one question asked four times: what is this item for? Two of them are electrodes. An electrode is named by which way electrons move there. In means reduction, a gain of electrons. Out means oxidation, a loss. One item is the electrolyte sitting between the two, and it has a job of its own. The last is not part of the cell at all, so ask whether it settles a direction or an amount.

ASK 10W

Match each item on Achebe's board to what it does.

BEHIND THE BUTTON 207W

Why an electron transfer always has two halves. Each of these reactions is one substance losing electrons and another gaining them. The two always happen together. Nothing loses electrons with nobody there to take them. Oxidation is the losing half; reduction is the gaining half. A cell's trick is to make those two halves happen a few centimetres apart.

What the electrolyte does, and why it has to refuse electrons. Charge has to get back somehow or the cell stops in a moment. The electrolyte lets ions carry it — here oxide ions, O^{2-} , crossing the solid. What it will not carry is electrons. That refusal is the whole design. The electrons have no way home except the outside wire. That is where the work gets done, and where the current is measured.

Why the current setting is a different kind of thing. The two electrodes and the electrolyte decide what happens. The current decides how much. Faraday's law makes that exact. A fixed amount of charge is a fixed number of moles, because each molecule of product needs a set number of electrons. Two for a molecule of hydrogen. Four for one of oxygen. Charge is not a rough guide to yield. It is the count.

TAKES AS READ

oxidation is loss of electrons and reduction is gain of electrons • a current is a flow of charge • stoichiometry as a feed rate: turning a production target into kilograms in — taken as read

VERDICT 88W

Electrolysis splits one electron transfer into two places. At the cathode the supply pushes electrons in, so whatever is there gains them — that is reduction. At the anode electrons are pulled back out, so whatever is there loses them — that is oxidation. The electrolyte between carries ions across and refuses electrons, which forces the circuit through the wire and keeps the two products apart. The current

setting is different in kind: by Faraday's law, charge is moles, so it fixes the amount rather than the direction.

TAKEAWAY 22W

The two halves of one electron transfer can be run in separate places, so long as something carries an ion between them.

2.3 What has to be paid for

PLANT CONTROL · A MODULE · BALLPARK

SCENE 36W

Yusuf Demir, the life support officer, has the sol's energy ledger open. The reactor releases heat while it runs, while electrolysis takes most of the array output. He wants the difference stated plainly in the log.

GUIDE 89W

Demir wants the direction written down as a number, and the number is the free-energy change: $\Delta G = \Delta H - T\Delta S$. Splitting water has a known enthalpy change, ΔH . That is the heat it takes in. It has a known entropy change too, ΔS , which is how much more spread out things end up. A positive ΔG means the change will not run on its own. One warning before you place the tiles. One is in joules and the others in kilojoules, so it wants converting rather than adding.

ASK 25W

The free-energy change says whether a reaction runs on its own. What is it for splitting a mole of water at the cell's working temperature?

BEHIND THE BUTTON 164W

What the two terms are doing. The enthalpy term, ΔH , is the heat the change takes in or gives out. The entropy term, ΔS , is how much more spread out the matter and its energy end up — and it is multiplied by the temperature. That multiplication is why heating something can turn a change that will not run into one that will.

Why a positive answer is a bill. A positive free-energy change means the products sit higher than the reactants, so the energy has to come from outside the reaction. Here it comes from the solar array, through the cell, at a price in kilowatt-hours that the ledger on Demir's desk has to carry.

Why the reverse is free. Making methane is water splitting run backwards, and it gives out what the splitting takes in. That is one number with two signs, not two separate facts. It is why this plant can run one step on its own and never the other.

TAKES AS READ

some reactions give out heat and others take it in · electrical energy can be counted in kilowatt-hours

VERDICT 131W

Direction is set by the free-energy change: $\Delta G = \Delta H - T\Delta S$. Splitting a mole of liquid water takes in 286 kJ, so ΔH is positive. It makes two gases out of a liquid, so things end up more spread out. ΔS is positive too, at 163 J per kelvin per mole. At 298 K the $T\Delta S$ term is $298 \times 0.163 = 48.6$ kJ. That leaves $\Delta G = 286 - 48.6 \approx +237$ kJ per mole. Positive means the products sit higher than the reactants, and no change of equipment moves that. The energy has to arrive from outside, which is what the array is for. Making methane is the same number with the sign flipped. That is why one step runs on its own and the other never will.

TAKEAWAY 23W

Whether a change runs on its own and how fast it goes are two separate questions, and one of them has a bill.

2.4 Worth digging, or worth leaving

WATER PLANT · A MODULE · BELT

SCENE 42W

The hopper takes one charge at a time and the digging arm has brought in more than it can process. Some of what is out there gives up water when it is heated. The rest costs the same power and returns nothing.

GUIDE 73W

You have more material than the hopper can take, so the whole job is deciding what is worth heating. One thing decides it: whether heat gets water back out of the sample. Hydrated minerals will, because they have water locked into the crystal itself, and so will buried ice. Dry basaltic dust will not, however promising it looks. Judge each sample on what it gives up, not on where the arm found it.

ASK 16W

Send each sample to the bin that says whether heat gets water back out of it.

BEHIND THE BUTTON 176W

Why this is the plant's first decision. Every charge costs the same heating energy whatever is in it, and energy is the thing this station has least of. A hopper charged with dry dust has spent a sol's margin on nothing.

What heating a hydrated mineral actually does. The water in these minerals is not damp. It is built into the crystal, held in place as part of the structure. Heat breaks it loose and it leaves as vapour. The process is called dehydration, and it needs a few hundred degrees rather than a warm morning. What is left behind is the same mineral with its water gone.

What holds water here, and in what way. Hydrated sulphates and clays hold it chemically — water molecules built into the crystal, which come off at a few hundred degrees. Buried ice holds it as ice, and gives it up far more cheaply.

Weathered basalt, volcanic glass and windblown fines are anhydrous: no water in the structure to release. They are also most of what the arm reaches.

TAKES AS READ

some minerals hold water that heating can drive off · intermolecular forces, and why they set a boiling point — taken as read

VERDICT 153W

Water is the input the whole plant runs on, and heating is how it is got back. So the only question worth asking of a sample is whether heat gets water out of it.

Hydrated sulphates and clays say yes. The water is built into the crystal and comes off at a few hundred degrees. Buried ice says yes more cheaply still. Weathered basalt, volcanic glass and the windblown fines are anhydrous — no water in the structure at all. They also cover most of the site, and they look exactly like the material that does hold some. That is why the sorting happens out in the yard and not at the hopper. The cost of getting it wrong is not the sample. It is the charge.

The heater draws the same power whatever is in it. A hopper of dry dust has spent a sol's margin making steam that was never there.

TAKEAWAY 15W

Every charge costs the same energy, so what is in it is the whole decision.

END OF THE SOL 18W

A loop that recycles can afford a poor pass, and a plant with no hydrogen cannot afford anything.

The line that froze shut

PLAN CARD 63W

Sol 293. Three hours after the hot run began, methane stopped reaching the transfer tank. The refrigerator still makes its set point and the valve command says open. Something inside the line is closing the passage. Today you use pressure, temperature and flow to decide what froze, then trace that material back through the plant to the water system that let it through.

BRIEFING 22W

The cold end stops passing liquid, turning yesterday's production gain into a blockage that has to be diagnosed from outside the pipe.

WHAT YOU DECIDE 16W

Find what closed the cold line, and what the hot run has to do with it.

3.1 Reading a blockage from outside the pipe

COLD END • A MODULE • DIAGNOSIS

SCENE 33W

Five readings are up on the cold end panel. Two are alarming and three are entirely ordinary. Cho says the ordinary ones are the reason most of the obvious answers do not survive.

GUIDE 95W

Cho's point is the one worth taking. Two readings are shouting and three are perfectly ordinary, and it is the ordinary three that will do the work. A reading that has not moved is a positive result — it rules out every explanation that would have moved it. So take each candidate in turn and ask how many of the five it actually fits. And look at the shape of the flow trace, not only its value. A flow that dies away over three hours is not telling the same story as one that stops.

ASK 9W

Which explanation fits every reading, not the loudest one?

BEHIND THE BUTTON 201W

Why water is the suspect and not the methane. The cold line falls from about 10 degrees to 115 kelvin. Water turns solid at its process frost/freezing condition, right at the top of that run. Methane does not become liquid until about the methane condenser's defined condition, at the very bottom. So water that gets past the drier has hundreds of degrees of pipe to freeze in. All of it comes before the methane end.

Why the unremarkable readings decide it. The alarming reading draws the eye, and it usually fits several explanations at once, so it rarely settles anything. The readings that separate candidates are the ones an explanation says should have moved and which have not. A normal value is a positive result against every mechanism that would have disturbed it.

How to work the four without fooling yourself. Take each one and predict the panel it implies before you look again — which readings it drives, which way, and roughly how far. Then compare. Several will fit part of the panel, deliberately, and a partial fit is exactly what a confident wrong answer feels like from inside. When two are left, find the reading their predictions disagree on.

TAKES AS READ

a refrigerator that has reached its set point is doing its job • water freezes at a far higher temperature than methane condenses at • stoichiometry as a feed rate: turning a production target into kilograms in — taken as read

VERDICT 99W

The quiet readings do the work. The cold head is sitting at its set point, the temperature it is asked to hold. So the refrigerator has not lost capacity. Jacket pressure is steady, so the jacket has not failed. That leaves something blocking the line. The shape of the flow trace says which kind. A shut valve stops the flow in a second. A solid builds up over hours and takes the flow down with it. The drier

outlet at 900 ppm — parts per million — names the material. Rising pressure upstream of a growing plug finishes it.

TAKEAWAY 26W

A blockage is diagnosed from the pattern of pressure, flow and temperature; the material is inferred from where in this process it could freeze or frost.

3.2 Why the hotter pass gave less

REACTOR HALL · A PERSON · BALLPARK

SCENE 31W

The hot run's assay is on Herrera's desk. Conversion per pass fell from 71% to 64%. Throughput rose. He wants the reason written down before anyone argues for another temperature increase.

GUIDE 88W

Herrera wants this settled with a number instead of an argument. The assay gives him one. The reaction quotient Q is built exactly like the equilibrium constant K . Each gas's partial pressure goes in raised to its coefficient in the balanced equation. Products over reactants. The difference is that Q is worked out at the mixture's own conditions. Wherever it happens to be right now. Compare Q with K and you have the direction. Two tiles are the raw pressures with no exponent applied. So read the labels.

ASK 21W

Work out the reaction quotient Q at the hot run's outlet, and say which way that mixture still has to go.

BEHIND THE BUTTON 170W

What Q is, and what it is not. Q is built exactly like K . But it is worked out wherever the mixture happens to be. So it describes the present. Not a fixed property of the reaction. Comparing the two tells you one thing. Which way the reaction still has to go. It tells you nothing about how fast it will get there. That is a different question with a different answer.

Why the exponents are not decoration. Every partial pressure goes in raised to its coefficient from the balanced equation. Hydrogen's is four. Leave that out and Q changes by two orders of magnitude here. One mixture is already past its ceiling. The other is nowhere near it. That is the difference.

Why the hot run had two things wrong with it. Temperature moved the ceiling. The higher throughput moved the contact time. One assay cannot separate them. What it can settle is the ceiling. K at the hotter temperature is smaller. This outlet mixture is already above it.

TAKES AS READ

the equilibrium constant is fixed at a given temperature · the Sabatier reaction gives out heat

VERDICT 129W

The Sabatier reaction is $\text{CO}_2 + 4\text{H}_2 \rightarrow \text{CH}_4 + 2\text{H}_2\text{O}$. So $Q = p(\text{CH}_4) \times p(\text{H}_2\text{O})^2 \div (p(\text{CO}_2) \times p(\text{H}_2)^4)$. At the hot outlet that is $0.42 \times 0.7056 \div (0.10 \times 0.0105) \approx 282$. K is 192 at 573 K. It is about 1,350 at 533 K. Smaller when hotter, because this reaction gives off heat. So at the hot setting the outlet mixture sits above the ceiling. Q is bigger than K . The reactor would have to run backwards to reach balance. At the cooler setting it is comfortably below. That settles the thermodynamic half of

Herrera's question. Not the kinetic half. The higher throughput also cut the contact time. Only a flow test can separate those two. It has to hold temperature and pressure fixed.

TAKEAWAY 21W

Heating an exothermic reaction lowers its equilibrium product yield. The observed reactor conversion can also depend on kinetics and residence time.

3.3 Getting water out of frozen ground

WATER PLANT · A MODULE · SEQUENCE

SCENE 33W

Femi Abiola, the regolith and water lead, has the plant's four steps written on cards. The morning crew disagreed about where the extra water appeared. He wants the cards restored to process order.

GUIDE 90W

Abiola's four cards will all happen; the only question is what each one needs from the step above it, so ask that of each. Ice that melts loose in the bucket runs straight into the ground. Vapour let go outside a sealed vessel joins an atmosphere at six millibars and is gone. And an ion-exchange bed — the column that swaps unwanted ions out of the water and onto a resin — can only be fed liquid. Put them out of order and each step finds nothing to work on.

ASK 8W

Order the steps the water plant runs in.

BEHIND THE BUTTON 201W

Why the water leaves as vapour and not as liquid. At six millibars — under a hundredth of the pressure at sea level on Earth — water has almost no liquid range at all. Warm ice-bearing ground in the open and the ice goes straight to vapour and disperses. That is why the hopper is sealed before it is warmed. The same warming then has nowhere to send the vapour except onto the chilled plate.

Why the column comes last and not first. The polishing column swaps unwanted ions — chloride, here — out of the water and onto a resin. What it has is a fixed capacity, not a percentage. Feed it the whole soil load instead of the water that load gave up, and it fills in a sol. It sits at the outlet of the holding tank because that is where the volume is smallest and already cleanest.

Why the order is graded whole. A sequence is a claim about what depends on what: each step is here because the one before it has already happened. One swapped pair breaks that claim wherever it sits, so partial credit would be credit for a plant that does not run.

TAKES AS READ

warming ice-bearing soil releases water as vapour at low pressure · dissolved salts stay behind when water evaporates · intermolecular forces, and why they set a boiling point — taken as read

VERDICT 87W

The order is set by what each step needs from the one above it. Ice that melts in the bucket runs into the ground and is lost, so the load stays frozen until it is inside something sealed. Vapour has to be caught where it is released, or it joins an atmosphere at six millibars and is gone. Condensing gives liquid water, and only liquid water can be run through an ion-exchange bed. Take the steps out of order and each one finds nothing to work on.

TAKEAWAY 14W

Every step in the water plant fixes something the step after it cannot survive.

3.4 Rate is not yield — Review

CATALYST BAY · A MODULE · CALLBACK · CHOICE

SCENE 29W

The greenhouse reactor is running cold and slow. Its operator proposes 40 degrees more, and the same argument that ran through the propellant plant a fortnight ago starts again.

GUIDE 66W

The same argument as the propellant plant, in a different building. Each option here makes two claims at once, so check both halves separately. One is about how fast the reaction gets to its balance. The other is about how much product that balance holds. Those are two different quantities, and for a reaction that gives off heat they are free to disagree with each other.

ASK 15W

A reaction that gives out heat is run hotter. Which pair of statements is right?

BEHIND THE BUTTON 193W

Why one setting moves two quantities. Temperature reaches the reaction twice over. It reaches the rate constant, which climbs steeply because more collisions arrive carrying enough energy to react. And it reaches the equilibrium constant, which for a heat-releasing reaction falls, because heat behaves as one of the products. Two separate routes, and nothing says they have to agree.

Why this is the normal case, not a special one. It is tempting to expect a single setting to have a single effect. Most of the interesting decisions in a plant are the other kind. Heat helps the rate and hurts the ceiling. More current makes more hydrogen and more waste heat. Faster flow raises throughput and cuts contact time. Every one of those is one knob pulling two ways.

What the industry actually does about it. If heat buys speed and costs yield, the usual answer is to buy the speed some other way. Run the bed at the coolest temperature that gives an acceptable balance. Then spend money on a catalyst good enough to reach that balance in the time available. That is why catalyst research is worth what it is worth.

TAKES AS READ

heating raises the fraction of collisions that clear the barrier · an exothermic reaction's balance is pushed back by heating

VERDICT 90W

Temperature reaches this reaction by two routes, and they run opposite ways. The rate constant climbs steeply, because more collisions arrive carrying enough energy to react. The equilibrium constant falls, because heat behaves as one of the products of a reaction that gives heat off. So the bed gets to its balance faster and the balance holds less. Every exothermic industrial reaction lives with that trade. The usual answer is a temperature low enough for a decent balance, and a catalyst good enough to reach it in the time available.

TAKEAWAY 17W

One setting moving two quantities in opposite directions is the normal case rather than a special one.

END OF THE SOL 14W

A plant is a chain, and the fastest step decides nothing on its own.

The feed that caps the plant

PLAN CARD 63W

Sol 294. Liquid is flowing again, but the plant cannot make methane from a feed it does not have. Today you turn the methane target into a carbon-dioxide requirement, audit where electrolysis charge is being lost, and follow every dissolved species in the recovered water to the equipment it can damage. The bottleneck is now a supply chain, not a reactor set point.

BRIEFING 23W

With the cold line clear, the question becomes whether carbon dioxide, hydrogen, or clean water is the feed that actually sets today's ceiling.

WHAT YOU DECIDE 13W

Size the feed the plant needs, and find where the current is going.

4.1 How much gas the bed has to be fed

REACTOR HALL • A MODULE • BALLPARK

SCENE 31W

The plant has to average 13.8 kilograms of methane a sol from here to the window. Herrera wants the carbon dioxide that implies before the intake compressors are asked for anything.

GUIDE 84W

Herrera has a production target and wants the feed it implies. This is stoichiometry, which is mole bookkeeping. The balanced equation does the work. One carbon atom leaves each carbon dioxide and arrives in one methane. So by moles the two go one for one. By mass they do not, because a carbon dioxide molecule is heavier than a methane one. Five numbers are on the board and two belong to other parts of the equation. Ask of each whether carbon bookkeeping needs it.

ASK 11W

Estimate the carbon dioxide the reactor must be fed each sol.

BEHIND THE BUTTON 174W

What a mole is doing in this sum. A mole is a fixed count of molecules. Think of the chemist's dozen, only very much larger. Balanced equations are written in moles because reactions happen molecule by molecule. So the equation says one carbon dioxide gives one methane, whether you are handling grams or tonnes.

Why the mass ratio is not one to one. Molar mass is what one mole of a substance weighs: 44 grams for carbon dioxide, 16 for methane. Same count of molecules, very different weight, because each carbon dioxide carries two oxygen atoms that the methane does not keep. So equal moles come out as 44 kilograms in for every 16 kilograms out.

Why the tiles carry labels. A bare number cannot be checked against the relationship it is going into. Reading the label catches a unit that does not match. It also catches a quantity that belongs to a different part of the problem. Do it before you place the tile. That habit is the whole point of this format.

TAKES AS READ

one mole of carbon dioxide yields one mole of methane in this reaction • a molar mass converts moles to kilograms

VERDICT 86W

Stoichiometry is mole bookkeeping. The balanced equation sends one carbon atom out of each carbon-dioxide molecule and into one methane molecule. So by moles the two go one for one. Mass needs one more step. A mole of carbon dioxide weighs 44 grams; a mole of methane only 16. Every kilogram of methane therefore needs 44/16 kilograms of carbon dioxide in. The intake compressors gather that out of a very thin atmosphere. The work they do scales straight off the production target the plant has promised.

TAKEAWAY 20W

Stoichiometry turns a production target into a feed rate, and the feed rate is what the intake has to meet.

4.2 Where 8% of the charge went

ELECTROLYSIS HALL · A PERSON · BALANCE

SCENE 35W

The cell held 640 amps for a full sol. Faraday's law predicts about 594 grams of hydrogen, but the separator collected only 546. Achebe has already checked the collection-line pressure and the hydrogen inventory downstream.

GUIDE 70W

Achebe has done the sensible checks already, so start where she left off. Faraday's law turns charge into a mass of product. 640 amps held for a sol should give about 594 grams of hydrogen. The separator collected 546. She has cleared the collection line and the inventory downstream, so nothing is hiding in a pipe. Read every stream on the ledger. Then count only what genuinely belongs in it.

ASK 15W

What term closes the charge ledger, and what does that term measure about the cell?

BEHIND THE BUTTON 181W

What Faraday's law claims. Every electron that crosses the cell makes a fixed amount of hydrogen — two electrons for each molecule of H_2 . So the predicted mass is arithmetic on the charge. It is an upper bound rather than a forecast. It assumes every single electron did the job it was sent to do.

Why the gap is a measurement and not an error. Electrons that did something else still crossed, and still counted as charge. Some drove a side reaction at the electrode. Some ran as a shuttle current, carrying charge back and forth to no purpose. Some made hydrogen that recombined inside the cell before it ever reached the separator. What is left over measures how much of the current did useful work. That fraction has a name: current efficiency.

Why checking the line first was right. A leak would explain the whole gap. So would hydrogen standing in the collection line that was not standing there yesterday. Neither says anything about the cell. Ruling both out is what turns an awkward discrepancy into evidence about the electrochemistry.

TAKES AS READ

Faraday's law converts charge into a theoretical amount of product · current efficiency is collected product divided by theoretical product from charge

VERDICT 88W

Current times time gives charge. Faraday's law turns that charge into a mass of product, because each electron makes a fixed amount of hydrogen. This run's charge is worth about 594 grams. Only 546 were collected. The line inventory and the measured vent loss both sat still, so the missing 48 grams are not hydrogen hiding downstream. They are charge that crossed the cell and did something other than make collectable hydrogen. $546/594$ is about 92 per cent. That ratio is the cell's current efficiency for the run.

TAKEAWAY 20W

Faraday's law fixes the theoretical product from charge; current efficiency measures how much of that charge reached the intended reaction.

4.3 What comes with the water

WATER PLANT • A MODULE • PROTOCOL

SCENE 35W

The holding tank's analysis is on Abiola's bench: four dissolved species, none of them water. Each one has somewhere downstream that it matters, and the plant has one polishing column between them and everything else.

GUIDE 73W

Abiola has four dissolved species on the bench and one polishing column to spend. So for each one, ask where it ends up and what it touches when it gets there. A catalyst's nickel surface? The hot faces inside a cell, where something can come back out of solution as scale? A packed bed it can physically clog? Or somewhere the plant does not much mind, because not every impurity is equally dangerous.

ASK 9W

Match each dissolved species to what it does downstream.

BEHIND THE BUTTON 150W

Why dissolved salts are not interchangeable. Martian brine can contain perchlorate, chloride and sulphate among other ions. Perchlorate ClO_4^- is not chloride Cl^- ; dissolving one does not turn it into the other. Each has to be tracked separately.

Why chloride is the warning species in this authored plant. Halide breakthrough into the electrolysis feed can create trace chlorine-bearing carryover downstream. Nickel methanation catalysts are sensitive to chlorine/halide poisoning, so a rising halide signal is treated as a catalyst-protection alarm rather than as proof that liquid chloride itself rode unchanged into the gas reactor.

Why the harmless one is harmless. Dissolved carbon dioxide makes the water slightly acidic and then arrives at a reactor whose entire job is turning carbon dioxide into methane. It is already on the feed list. Worth noticing, because the instinct to remove everything measurable spends a polishing column's fixed capacity on something the plant wanted anyway.

TAKES AS READ

salts dissolve into ions in water • an electrolysis cell and a catalyst bed both have surfaces that can be fouled • intermolecular forces, and why they set a boiling point — taken as read

VERDICT 77W

A dissolved species matters because of where it can travel and what it can damage. Chloride is distinct from perchlorate, and this plant treats halide breakthrough as a catalyst-protection signal because chlorine-bearing carryover can poison nickel. Sulphate can form scale on warm process surfaces. Fine particles can load filters and driers. Dissolved carbon dioxide is already part of the downstream carbon feed. The polishing train is therefore designed around separate consequences, not a single idea of 'salt'.

TAKEAWAY 19W

The brine contains different ions; track each one by its downstream consequence instead of treating 'salt' as one contaminant.

4.4 Three hundred degrees, while the feed wanders

REACTOR HALL · A MODULE · HOLD

SCENE 36W

The Sabatier bed has to sit at its working temperature while the electrolysis stacks feed it. The stacks follow the array, the array follows the sun, and the bed heats itself whenever it is making methane.

GUIDE 77W

You are holding the bed at its working temperature, inside the band on the panel. Watch what the band does. It narrows as the sol goes on, because the tank has a filling rate to make. A bed wandering ten degrees is not making it. Your control is the cooling-loop valve. One thing to carry in with you: the bed heats itself whenever it is working. This is not a temperature that stays put on its own.

ASK 10W

Hold the bed at its working temperature through the sol.

BEHIND THE BUTTON 177W

Why the bed heats itself. Assembling methane gives off heat, so a bed that is working is a bed that is warming. And a bed that warms converts less on each pass, because the ceiling falls as the temperature rises. The reaction and the cooling are in a loop with each other. Neither settles without the other.

Why the feed will not sit still. The stacks run on array power, the array follows the sun and the dust, and more hydrogen arriving means more reaction, which means more heat. It arrives as a rate rather than as a step, which is what makes chasing the reading a losing game.

What holding a self-heating process asks of you. A disturbance that keeps pushing needs the valve set to a new position, not nudged and watched. Nudge, wait, nudge again, and the bed is always a few minutes behind the feed. By the time the reading has moved far enough to convince you, the heat that moved it is already in the metal. Answer the feed, not the reading.

TAKES AS READ

a reaction that gives out heat warms what it is running in

VERDICT 147W

The Sabatier reaction gives off heat, so the bed warms whenever it is working. A warmer bed has a lower ceiling. That is why this plant treats temperature as a production setting rather than a comfort one. Every disturbance here is a rate, not a step. More hydrogen from the stacks does not lift the bed five degrees and stop. It raises the reaction rate, which raises the heat coming out. That goes on lifting the bed until the cooling is set to carry it away. So the valve is moved to a new position and left there. The band narrows for a reason. The tank has a filling schedule and the vehicle has a launch window. A bed wandering ten degrees either way is not making the rate the plan was built on. By late sol there is no margin left to make it up in.

TAKEAWAY 13W

A process that heats itself is held by whoever is watching the cooling.

END OF THE SOL 20W

A plant is limited by whichever feed runs out first, and it will not be the same one every sol.

The ceiling on heat

BEFORE THE SOL — TRIAL

The rover, and the three hundred metres to the ice cut

The water this plant runs on is cut out of the ground 338 metres north-west, and until today nobody has asked you to go there. It is too far to walk in a suit and get any work done afterwards, which is what the rovers are for and why they are signed out this morning. Drive it once while the sky is clear, so you know the track before you need it in a storm.

PLAN CARD 68W

Sol 295. The first hot run helped because the bed was short of time. Now another 40 °C is on the board. Herrera says the plant is close enough to equilibrium that more heat will lower the methane ceiling; Sundqvist says time still matters. Today you put numbers on the thermodynamic limit, state what the catalyst can change, and rebuild the cold-end procedure so yesterday's blockage cannot recur.

BRIEFING 22W

Another temperature increase is proposed, and this time the plant has enough evidence to separate a faster reaction from a better equilibrium.

WHAT YOU DECIDE 12W

Put a number on how far the temperature argument can be taken.

5.1 The temperature the reaction stops wanting to run

PLANT CONTROL • A MODULE • BALLPARK

SCENE 43W

Demir has the plant's simplified thermodynamic table for the Sabatier loop: $\Delta H = -123.6$ kJ/mol and $\Delta S = -172$ J/mol/K over the training range. He wants the temperature where this approximation makes ΔG cross zero, knowing the real plant uses temperature-dependent property data.

GUIDE 90W

Demir has the data sheet and wants ΔG at the bed's own temperature. So this is $\Delta G = \Delta H - T\Delta S$ with real numbers in it. Both terms here happen to be negative, and they pull against each other once the temperature multiplies the second one. That is the interesting part. As the bed gets hotter the second term grows, so somewhere up the scale the whole thing changes sign. Five numbers are on the board and two belong to other calculations. Ask of each whether this one needs it.

ASK 11W

Estimate the free-energy change for the reaction at the bed's temperature.

BEHIND THE BUTTON 194W

Why both terms are negative here, and why that matters. Making methane gives off heat, so ΔH is negative and the reaction is favoured on those grounds. But five gas molecules go in and three come out. The mixture ends up less spread out than it started, so ΔS is negative too. Subtracting a negative $T\Delta S$ adds a positive amount. Entropy is working against this reaction, not for it.

Why the temperature decides which term wins. ΔH does not care how hot the bed is. $T\Delta S$ does, because T is a multiplier sitting right on it. So heating the bed grows the term that opposes the reaction while leaving the term that favours it alone. Push far enough and ΔG crosses zero — with these tabulated numbers, near 720 K. Above that the reaction no longer runs on its own at all.

Why the tiles carry labels. A bare number cannot be checked against the relationship it is going into. Reading the label is how you catch a unit that does not match, or a quantity belonging to a different problem, before you place it. That habit is the whole point of this format.

TAKES AS READ

free energy combines the heat of a reaction with its entropy change • a negative free-energy change means the reaction can run on its own

VERDICT 114W

The two terms pull against each other. Making methane gives off heat, so ΔH is negative and that favours the products. But it also turns five gas molecules into three. So ΔS is negative as well, and a negative ΔS works against the reaction once the temperature multiplies it. ΔH does not change with temperature; $T\Delta S$ does. So the hotter the bed, the more the opposing term weighs. At 593 K, $\Delta G = -123,600 - (593 \times -172) = -21,600$ J per mole. Still comfortably negative. The same

tabulated numbers cross zero near 720 K. Real values drift with temperature, so treat 720 K as this course's threshold rather than an exact plant limit.

TAKEAWAY 18W

A simplified ΔG model can locate a thermodynamic trend; real reactor design uses temperature-dependent data and measured equilibrium.

5.2 What a catalyst cannot do

CATALYST BAY · A PERSON · CHOICE

SCENE 36W

Kaur has two charges on the bench: the spent one from sol 288 and a fresh one still in its sealed can. Sundqvist's proposal assumes a better catalyst would raise the methane a pass can give.

GUIDE 72W

Sundqvist's proposal rests on a claim about catalysts, and it is worth testing the claim before the proposal. A catalyst lowers the activation energy — the push a collision needs before it reacts. Hold on to one detail while you read the four options. It lowers that push in both directions at once. Forwards and backwards, by the same amount. Then ask of each option whether a catalyst could actually deliver it.

ASK 13W

A better catalyst is fitted at the same temperature and pressure. What changes?

BEHIND THE BUTTON 214W

What a catalyst is actually doing. Molecules only react when a collision carries enough energy to get over a hump — the activation energy. A catalyst gives the reaction an easier route over that hump, usually by holding the molecules on a surface in a helpful arrangement. It is not consumed, and it does not appear in the balanced equation. It changes the road, not the destination.

Why it cannot move the balance point. The equilibrium constant K is set by the ratio of the forward speed to the backward speed. An easier route is easier in both directions at once, so both speeds rise and their ratio does not move. That ratio is K . This is the part people reliably get wrong, because it feels as though anything that helps a reaction should give you more of the product.

Why it cannot change the heat either. The heat a reaction gives out depends on two things only: what the molecules were at the start, and what they are at the end. It is the energy difference between them. The road taken in between does not enter into it. So a catalyst buys time and nothing else. In a plant with a fixed number of sols left, that is still worth a great deal.

TAKES AS READ

a catalyst offers a reaction an easier path · equilibrium is where the forward and reverse rates match

VERDICT 94W

A catalyst lowers the activation energy, and it lowers it by the same amount in both directions. Forward and backward both speed up. So the ratio between them does not move, and that ratio is the equilibrium constant. Nor does a catalyst change how much heat comes out. That depends on what the molecules are at each end, not on the road taken between them. So a better catalyst buys time and no extra methane at all. In a plant with a fixed number of sols left, time is still worth a great deal.

TAKEAWAY 17W

A catalyst changes how long the balance takes to arrive and leaves the balance where it was.

5.3 Set the cold end in the only safe order

COLD END · A MODULE · SEQUENCE

SCENE 36W

Cho is rebuilding the cold end's written procedure after the blockage, and wants the four stages back in the order the stream meets them. 2 of the 4 exist because of what happened on sol 293.

GUIDE 51W

The blockage taught the plant that temperature alone is not the procedure. Each phase boundary depends on pressure and composition. Put the steps in the order that removes water and carbon dioxide before deep cooling, then set the pressure for the methane condenser before using its roughly 112 K normal-boiling-point scale.

ASK 11W

Order the cold-end steps from dirty warm gas to liquid methane.

BEHIND THE BUTTON 101W

Why water comes out first. Residual water is removed while the stream is still warm enough that it cannot freeze onto cryogenic metal. Letting it reach the deep-cold section recreates Sol 293.

Why carbon dioxide has its own trap. Its frost point depends on its partial pressure, so the plant removes it in a dedicated adsorber/trap instead of pretending 195 K is a universal switch.

Why pressure is part of the methane temperature. Methane boils near 111.7 K at one atmosphere. The stream therefore has to be at the condenser's defined pressure before a temperature such as 112 K means anything.

TAKES AS READ

water and residual carbon dioxide are removed before the deepest cryogenic stage · the final methane condenser is operated near 1 bar · intermolecular forces, and why they set a boiling point — taken as read

VERDICT 68W

First remove residual water at a warm drying stage. Then remove carbon dioxide in its dedicated trap before deep cooling. Only the cleaned methane-rich stream is let down to the condenser's defined pressure, near one atmosphere here. At that pressure methane's normal condensation/boiling temperature is about 112 K, so the final cooler can liquefy it without asking a dirty stream to cross every frost boundary inside one pipe.

TAKEAWAY 20W

Phase temperatures are conditional on pressure; the safe sequence removes contaminants first, then sets the methane condenser's pressure and temperature.

5.4 What the loop settles at — Review

REACTOR HALL • A MODULE • CALLBACK • PROTOCOL

SCENE 36W

A second reactor on the station makes ammonia for the greenhouse, out of nitrogen and hydrogen. It gives out heat, and it ends with fewer gas molecules than it started. Four changes are proposed to it.

GUIDE 67W

Different reactor, different product, same four questions. That is the point of putting it in front of you again. For pressure, count the gas molecules on each side. For temperature, ask which side the heat is on. For anything added or taken away, ask whether it takes part in the reaction at all. And for the catalyst, ask whether it changes the destination or only the journey.

ASK 12W

Match each change to what it does to the ammonia at equilibrium.

BEHIND THE BUTTON 194W

Why the substance in the pipe does not matter. Le Châtelier's principle is about the shape of the equation, not about what the molecules are. $\text{N}_2 + 3\text{H}_2 \rightleftharpoons 2\text{NH}_3$ has four gas molecules going in and two coming out, and it gives off heat. So does the Sabatier reaction, in its own numbers. Two different plants, two different products, and the same four answers.

Why counting molecules is the whole of the pressure question. Squeeze a mixture and it shifts to whichever side occupies less room, because that relieves some of the squeeze. Fewer gas molecules means less room. So the side with fewer wins. Here that is the ammonia side, four going to two, and it is why industrial ammonia plants run at pressures that sound absurd.

Why the catalyst is on this board at all. It is here to be the one that does not move the balance. Every other change on the list shifts where the mixture settles. The catalyst changes how long it takes to get there and leaves the destination exactly as it was. Meeting that fact in a second reactor is the point of meeting it twice.

TAKES AS READ

a reaction that gives out heat has its balance pushed back by heating • fewer gas molecules on one side makes that side favoured by pressure

VERDICT 92W

The shape of the equation decides all four answers. That is why a reactor making a completely different substance behaves the same way. Count the gas molecules and you know what pressure does: four in, two out, so squeezing favours the ammonia. Ask which side the heat is on and you know what temperature does: heat is an output here, so adding more pushes the balance back. Ask whether what was removed takes part, and you know whether the balance moves at all. And the catalyst changes the time, never the destination.

TAKEAWAY 16W

A rule about balance is about the equation's shape, not about the substance in the pipe.

END OF THE SOL 13W

An argument that has an arithmetic answer should not be settled by seniority.

The water bottleneck

PLAN CARD 66W

Sol 296. The drill is bringing in ground with only four percent water. The holding tank is saltier than anything the plant has used this season, and the electrolysis stacks are already power-hungry. Today you estimate what an ideal freezing-point model predicts, compare that with what the tank actually did overnight, and decide what the extra cell voltage means for the station's heat and power budget.

BRIEFING 22W

The shallow cut has fallen to four percent water, forcing the plant to distinguish measured brine behavior from an ideal chemistry model.

WHAT YOU DECIDE 14W

Find the feed that actually caps the plant, and what it costs to clean.

6.1 The brine the ideal model cannot finish

WATER PLANT · A MODULE · BALLPARK

SCENE 37W

The holding tank sat at minus 40 overnight. It drained as a liquid this morning. Abiola has the analysis. It is magnesium perchlorate, and a great deal of it. He wants the freezing point that concentration implies.

GUIDE 88W

This tank drained as a liquid at minus 40. That tells you something about how much salt is in it. The rule is freezing-point depression. $\Delta T_f = i \times K_f \times m$. Molality m is moles of salt per kilogram of water. K_f is a constant belonging to water itself. And i is the particle count. It is how many separate ions one formula unit breaks into. That count is what matters here. Not how heavy the salt is. Two of the five numbers belong to other questions.

ASK 16W

What freezing-point drop does the ideal dilute-solution model predict, and what can that number actually support?

BEHIND THE BUTTON 194W

Why dissolved particles lower a freezing point at all. Ice can only form one way. Water molecules line up into a crystal. Dissolved particles get in the way of that. They do not stop it. They make it happen at a lower temperature than it otherwise would. This is called a colligative property. It depends on how many particles are dissolved. Not on what they are.

Why the particle count is the whole trick. Magnesium perchlorate is $\text{Mg}(\text{ClO}_4)_2$. Put it in water and it does not stay whole. It breaks into one magnesium ion and two perchlorate ions. That is three separate particles from one formula unit. Something that dissolves in one piece gives one particle. So this salt depresses the freezing point three times as much. That factor of three is i , the van 't Hoff factor.

Why 45 K is a magnitude. Not a measurement. The rule assumes dissolved particles ignore each other. At 8 moles per kilogram they cannot. The tank is packed, and real ions attract and crowd. So treat 45 K as showing why this brine is still liquid at minus 40. It is not its true freezing point.

TAKES AS READ

a dissolved salt separates into ions in water · the freezing point of pure water is 273 K

VERDICT 93W

The ideal colligative model counts dissolved particles. Magnesium perchlorate contributes roughly three ions per formula unit in that simplified picture, giving $\Delta T_f = 3 \times 1.86 \times 8.0 \approx 45 \text{ K}$. But 8 mol/kg is far outside the dilute regime where that equation is quantitatively trustworthy: ion interactions and real phase equilibria matter enormously. So 45 K is a first-order scale, not the tank's certified freezing

point. The observation that the brine actually drained as liquid at $-40\text{ }^{\circ}\text{C}$ is a measurement; the calculation only shows why a very large depression is plausible.

TAKEAWAY 22W

At extreme concentration, the ideal colligative calculation gives a scale; the tank's measured phase behavior is the evidence the plant operates on.

6.2 Which part of 1.75 volts heats the stack?

ELECTROLYSIS HALL · A PERSON · CHOICE

SCENE 48W

The water stack has a reversible potential near 1.23 V at these reference conditions, a thermoneutral benchmark near 1.48 V, and it is operated at 1.75 V. Demir wants to know why subtracting 1.23 V and calling the whole remainder 'waste heat' is not the right energy story.

GUIDE 51W

Three voltage scales are on the board. About 1.23 V corresponds to the reversible Gibbs free-energy requirement. Around 1.48 V is the thermoneutral point where the electrical input matches the reaction enthalpy. Real kinetics and resistance push the operating voltage higher still. Separate those before deciding what becomes net stack heat.

ASK 12W

At 1.75 V, which interpretation of the cell's energy balance is defensible?

BEHIND THE BUTTON 101W

What 1.23 V means. It is the reversible electrical free-energy requirement under the stated reference conditions. It is not a promise that a practical cell can carry useful current there.

What the thermoneutral voltage adds. Splitting liquid water also has an enthalpy requirement. The difference between the reversible and thermoneutral scales can be supplied or removed as heat depending on cell conditions; it is not automatically electrical waste.

Where real losses appear. Electrode kinetics and ohmic resistance require extra driving voltage. At an operating voltage above the thermoneutral benchmark, those losses contribute to net heat that the cooling system must reject.

TAKES AS READ

the reference reversible voltage is about 1.23 V and thermoneutral voltage about 1.48 V for this simplified water-electrolysis comparison · real electrode and resistive losses require additional voltage at useful current

VERDICT 91W

The reversible 1.23 V scale corresponds to Gibbs free energy. The thermoneutral scale near 1.48 V accounts for the full reaction enthalpy under the simplified reference conditions. Real kinetics and resistance then push the operating cell to 1.75 V. So it is wrong to label every volt above 1.23 as waste heat. The cell's net heat balance depends on the thermoneutral benchmark, while kinetic and ohmic losses are the part that undeniably increase electrical cost and heat rejection. The hydrogen molecule itself is not 'higher energy' because the cell was inefficient.

TAKEAWAY 16W

Reversible free energy, reaction enthalpy and real cell losses are different parts of the energy balance.

6.3 Compute which way it runs

REACTOR HALL · A MODULE · BALLPARK

SCENE 37W

Herrera freezes the 09:40 gas-analyser screen before the loop has settled. Four partial pressures sit beside $K_p = 80$ for the plant's simplified 593 K equilibrium model. He wants the direction called before anyone changes a valve.

GUIDE 82W

Herrera has frozen the screen before the loop settled, which is exactly when Q is worth having. Build Q the same shape as K : each partial pressure raised to its coefficient, products over reactants. The difference is that Q uses the mixture on the screen right now. Then compare the two. Six numbers are on the board and two are not partial pressures at all. And be clear what the comparison gives you. It gives a direction. It says nothing about speed.

ASK 14W

Estimate Q from the four partial pressures. Is the mixture driven forward or backward?

BEHIND THE BUTTON 186W

How to read Q against K . If Q is smaller than K , the mixture is short of products, so the forward reaction wins until the two meet. If Q is bigger, there is too much product and the reaction runs backwards. If they are equal, the mixture is at equilibrium and nothing changes overall. Three cases, one comparison.

Why total pressure is not one of the numbers you want. Q is built from partial pressures — each gas's own share of the total. The total is the sum of those shares. Putting it into Q as well would be counting the same gas twice. It sits on the board because it is a real reading on a real gauge, not because the quotient wants it.

Why the answer is a direction and not a rate. Q against K tells you which way the mixture has to move. It says nothing about how long the move takes. A tired catalyst can leave a mixture far from equilibrium for hours. Temperature is the one setting that moves K itself; the catalyst only changes how fast the gap closes.

TAKES AS READ

the supplied $K_p = 80$ is the value used by this simplified loop model at 593 K · the gas snapshot is represented by the displayed partial pressures

VERDICT 79W

Q uses the same expression as K , applied to the mixture that exists right now. Here Q is about 0.016 and K is 80. So this mixture holds far too little product for equilibrium. The forward reaction wins, and goes on winning until Q has climbed to K . That is a direction, not a speed. A tired catalyst could take hours to close the gap. Temperature is the one setting that moves K itself; the catalyst cannot touch it.

TAKEAWAY 23W

Q versus K predicts the direction this model expects the mixture to move from the measured snapshot; it does not say how fast.

6.4 What freezes before methane condenses — Review

COLD END · A MODULE · CALLBACK · SEQUENCE

SCENE 30W

Cho has four species from a process stream and the phase boundaries calculated at the stated line pressure. She wants them ordered from the warmest removal condition to the coldest.

GUIDE 76W

All four are in the stream together, so nothing here happens before anything else. What sorts them is how strongly each one holds on to itself. So ask of each what kind of pull it has. Hydrogen bonds are the strongest of the three. Ordinary forces between whole molecules are weaker. Weakest of all is the fleeting attraction between electron clouds. And one of these is a single atom, with the least to hold on with.

ASK 14W

Order the substances by the temperature at which they leave the gas, warmest first.

BEHIND THE BUTTON 196W

Why attraction between molecules sets a boiling point. To leave a liquid, a molecule has to break away from its neighbours. The harder they hold on, the more energy that takes, and energy means temperature. So a substance whose molecules pull strongly on each other stays liquid to a high temperature. One whose molecules barely notice each other needs deep cold before it will condense at all.

The three strengths, in order. A hydrogen bond forms when hydrogen sits next to oxygen, nitrogen or fluorine. It is much the strongest of these three. That is why water is still liquid at temperatures where the others are all gas. Next come ordinary attractions between whole molecules. Weakest of all are the fleeting pulls between electron clouds, which every substance has and some substances have nothing else.

Why argon is last. It is a single atom, not a molecule. It has no shape and nothing sticking out. So nothing is available but that fleeting pull between electron clouds. And a small atom has a small cloud. It condenses near 87 K, colder than nitrogen, which is why recovering it means going deeper than anything else in this stream.

TAKES AS READ

each substance leaves the gas phase at its own temperature · stronger attractions between molecules mean a higher condensation temperature

VERDICT 57W

At the stated process pressure, the supplied phase-boundary data put water first, then carbon dioxide, then nitrogen and argon in deep cold. The important rule is not memorising four universal temperatures: condensation and deposition conditions move with pressure. The plant therefore calculates or measures the boundary for its own line before using temperature as a separation tool.

TAKEAWAY 15W

Phase boundaries can order a separation only after the pressure and composition have been stated.

END OF THE SOL 14W

The step that limits a plant is not the step anybody is arguing about.

The bed is dying

PLAN CARD 65W

Sol 297. The hottest point in the catalyst bed has moved 30 centimetres in nine sampled sols, and methane conversion is falling. Kaur says a working bed is becoming shorter; others blame flow, sensors or heat damage. Today you use the quiet readings to decide what is actually changing, while the tank farm turns a small heat leak into a measurable loss of flight oxygen.

BRIEFING 18W

The reactor still reaches its usual peak temperature, but the hot zone is walking downstream while conversion falls.

WHAT YOU DECIDE 12W

Find out what is happening inside a bed nobody can see into.

7.1 What the tank farm loses standing still

COLD END • A MODULE • BALLPARK

SCENE 39W

Élodie Moreau, the cryogenic fluids technician, logs the oxygen tank losing liquid on sols when nothing is drawn from it. The tank's insulation lets about 30 watts through, and she wants that expressed in kilograms rather than in watts.

GUIDE 80W

Moreau wants a leak stated in kilograms, so this is a heat sum turned into a mass. Here is the piece to hold on to. A liquid already sitting at its boiling point cannot get any warmer. So heat arriving is spent boiling it away instead. The energy each kilogram needs is the latent heat of vaporisation. Two of the five numbers belong to other questions. And notice that 30 watts sounds small but acts every second of the sol.

ASK 10W

Estimate the oxygen the tank loses to boil-off each sol.

BEHIND THE BUTTON 150W

Why this estimate can turn heat into mass. The tank is being treated as a pressure-controlled saturated cryogenic reservoir. Under that approximation the liquid remains near its saturation temperature and the dominant response to a small steady heat leak is vaporisation rather than a large temperature rise.

What latent heat of vaporisation means. It is the energy needed to turn one kilogram of a liquid into gas, with no change of temperature at all. For liquid oxygen it is 213,000 joules a kilogram. Divide the heat that arrived by that number and you have the kilograms that left.

Why 30 watts is not a small number. A watt is a joule every second, and a sol is 88,800 seconds long. So 30 watts delivers about 2.7 million joules a sol — enough to boil away 12.5 kg of oxygen. Nobody drew a drop from the tank. The insulation did it.

TAKES AS READ

for this first-order estimate the oxygen tank is held near a constant pressure by its pressure-control/vent system • the liquid is treated as saturated near 90 K and incoming heat is converted mainly into vaporisation • vapour pressure, and reading a cryogenic tank by it — taken as read

VERDICT 68W

Treat the pressure-controlled liquid-oxygen tank as saturated for this first-order estimate. Then the 30 W heat leak acts for 88,800 s and supplies about 2.66 MJ per sol. Dividing by the stated latent heat, 213 kJ/kg, gives about 12.5 kg of boil-off per sol. A real sealed or actively recondensed tank would need a fuller pressure-temperature model; this calculation is the loss scale under the stated operating assumption.

TAKEAWAY 14W

Heat reaching a cryogenic tank is paid for in kilograms rather than in degrees.

7.2 The reaction that likes being hot

PLANT CONTROL • A PERSON • CHOICE

SCENE 43W

Dana Stern, the analytical chemist, has carbon monoxide rising in the assay each time the reactor set point is raised. Methane falls by almost the same carbon amount. Kaur has the two candidate reactions on the board and wants the temperature effect explained.

GUIDE 80W

Kaur has two reactions on the board, not one, and they are competing for the same carbon. That is the thing to hold on to. Making methane gives off heat. Making carbon monoxide takes heat in. Ask of each option whether it keeps the two reactions separate, or treats the bed as doing one thing faster. What is changing here is not just how much converts. It is how the carbon gets divided up. The word for that is selectivity.

ASK 10W

Why does the hotter bed make relatively more carbon monoxide?

BEHIND THE BUTTON 176W

What selectivity means, and why it is a separate question from conversion. Conversion asks how much of the feed reacted. Selectivity asks what it turned into. A bed can convert more feed and still hand you less of what you wanted, if the extra went somewhere else. Two reactions sharing one carbon feed is exactly that situation.

Why heating helps one and hurts the other. Making methane gives off heat, so adding heat pushes that balance back. Making carbon monoxide takes heat in, so adding heat pulls that balance forward. Same bed, same set point, opposite effects. The two equilibrium constants simply do not move together. Endothermic and exothermic reactions respond to temperature in opposite directions, always.

Why speed is not the explanation. Both reactions do get faster as the bed warms. But suppose both sped up by the same sort of factor. The split between them would stay where it was, and the assay would read the same proportions. It does not. So the change is in the two balances, not in the two speeds.

TAKES AS READ

free energy combines an enthalpy term with a temperature times an entropy term • a reaction that takes in heat has a positive enthalpy change • rate constants, and how steeply they climb with temperature — taken as read

VERDICT 94W

Two reactions are competing for the same carbon, and they have to be judged separately. Making methane gives off heat, so heating pushes that balance back toward the feed. Making carbon monoxide takes heat in, so heating pulls that balance forward. Both get faster too, but speed alone would leave the proportions unchanged. What moved is the split. That split is selectivity, and it moved because

the two equilibrium constants do not travel in the same direction. So a set point does not only decide how much converts. It decides what the carbon becomes.

TAKEAWAY 14W

Temperature can change selectivity because competing reactions can have different thermodynamic responses to heating.

7.3 Reading a bed by where its heat is

CATALYST BAY · A MODULE · DIAGNOSIS

SCENE 40W

Nine sols of profiles are pinned along the wall in order. On the first, the hot spot sits 4 centimetres into the bed. On the last it sits at 34. The peak temperature is about the same on every one.

GUIDE 77W

Nine sols of profiles are on the wall and something is moving. But look at what is not moving. The peak temperature is the same on every one. The pressure drop across the bed has not changed. Neither has the feed. Those three are your evidence, not the hot spot. Take each candidate and ask how many of the five readings it fits. And ask one more thing: which end of a bed meets the strongest feed?

ASK 5W

Which explanation fits every reading?

BEHIND THE BUTTON 196W

Why a hot spot marks where the work is happening. The reaction gives off its heat exactly where it occurs. Dead catalyst does no reacting, so it gives off no heat — gas passes through it unchanged and unwarmed. The hot spot therefore sits at the first place the gas meets catalyst that still works. Move that boundary and the hot spot moves with it.

Why a bed dies from the inlet end. Whatever is poisoning this catalyst arrives dissolved in the feed. The front of the bed meets that feed first and at full strength, so it takes the damage first. As the front dies, the poison travels deeper before it is absorbed, and the dead zone grows downstream. A poisoned bed is eaten from the inlet, always in that direction.

Why the quiet readings decide this one. The alarming reading draws the eye and usually fits several explanations at once. The readings that separate them are the ones an explanation says should have moved and which have not. Unchanged pressure drop rules out packing and fouling. Unchanged feed rules out a leaner feed. An unchanged peak says the catalyst that still works is undamaged.

TAKES AS READ

a reaction gives out its heat where it is happening · a bed can lose activity in one part and keep it in another

VERDICT 92W

The unchanged peak temperature is what decides it. Wherever the reaction is happening, it is going as hard as ever. So the catalyst that still works has not been weakened. What has changed is how much of the bed is doing any work at all. Unchanged pressure drop rules out packing and fouling. Unchanged feed rules out a leaner feed reacting later. That leaves a dead front. A bed dies from the inlet end, because that is where the feed arrives at full strength, and conversion falls as the working length shrinks.

TAKEAWAY 11W

Where a bed is hot is where it is still working.

7.4 Which electrode does which job — Review

ELECTROLYSIS HALL • A MODULE • CALLBACK • PROTOCOL

SCENE 29W

The station's oxygen generator splits water in a low-temperature cell rather than a hot one. Achebe wants it read the same way as the hot stacks, part by part.

GUIDE 74W

A cold cell this time instead of a hot one, and the reading does not change. Ask of each part which way the electrons are being pushed. Electrons in means reduction, a gain. Electrons out means oxidation, a loss. That much is true of any water-splitting cell, whatever it is built from. One item on this board is not a place at all, so ask what a voltage does rather than where it is.

ASK 10W

Match each part of this cell to what happens there.

BEHIND THE BUTTON 111W

Why the electrode names do not depend on the cell. An electrode is named by the process occurring there: reduction where electrons are supplied, oxidation where electrons are removed.

What changes between the hot and cold cells. Different electrolytes carry different ions, but in both cases the electrolyte carries ionic charge while keeping the product gases separated from the external electron path.

Why voltage belongs on a board of parts. The reversible potential is the thermodynamic reference. A practical cell needs additional kinetic and resistive driving voltage to carry useful current, and the full heat balance also depends on the thermoneutral voltage rather than on one simple subtraction from 1.23 V.

TAKES AS READ

reduction happens where electrons are pushed in • the electrolyte carries ions and not electrons

VERDICT 86W

A cell is one redox reaction with its two halves held apart. Electrons supplied to an electrode drive reduction; electrons removed drive oxidation. The electrolyte carries ions rather than the external electrons and helps keep products separated. The applied voltage is a system condition, not an electrode: the reversible value is a thermodynamic reference, while practical operation requires additional kinetic and resistive driving voltage. That additional voltage increases electrical cost, but the complete heat balance should not be reduced to 'everything above the minimum becomes heat.'

TAKEAWAY 18W

The electrode roles follow electron transfer; practical voltage and heat depend on the cell's thermodynamics, kinetics and resistance.

A measurement taken every sol becomes evidence when somebody lays nine of them side by side.

The compound nobody asked for

BEFORE THE SOL — FOLLOW

Cho walks the cold end once

The liquefier's cold end is the part of the plant nobody has a diagram of at the level that matters — which joint frosts first, which line ticks when it is cooling. Cho, the cryogenics engineer, walks it once. Stay with her and keep off the lines; a boot on a cold transfer line is a leak you cannot weld out here.

PLAN CARD 62W

Sol 298. Stern has run the first full product assay since spring. One carbon-bearing compound has been present without a line on the routine report. At the same time, the water-polishing bed is approaching breakthrough. Today you turn current into a theoretical hydrogen ceiling, distinguish that ceiling from what is collected, and ask what a measurement misses when nobody asks the question.

BRIEFING 23W

A full product assay finds a compound the routine panel never looked for, while the water column shows the first signs of exhaustion.

WHAT YOU DECIDE 13W

Turn current into kilograms, and find out what is actually in the product.

8.1 From amps to kilograms

ELECTROLYSIS HALL · A MODULE · BALLPARK

SCENE 32W

1 cell in the water stack carried 640 amps for the whole sol. Achebe wants the hydrogen that represents, worked out from the charge rather than from what the collection line reported.

GUIDE 98W

Achebe wants the hydrogen worked out from the charge, not from what the collection line reported. That is Faraday's law, and it is bookkeeping. Current times time gives charge, in coulombs. Divide by the charge one mole of product needs and you have moles. Multiply by the molar mass and you have grams. Five numbers are on the board, and one is the cell voltage, which belongs to the energy bill rather than to the amount. And note what this answer assumes: every electron doing the job it was sent to do. The real figure will be lower.

ASK 10W

Estimate the hydrogen one cell should produce over a sol.

BEHIND THE BUTTON 169W

What Faraday's law actually says. Charge is counted in coulombs, and one mole of electrons carries 96,485 of them. That number is the Faraday constant. Making one molecule of hydrogen takes two electrons, so one mole of hydrogen takes two moles of them: 192,970 coulombs. Charge and moles are the same measurement in different units.

Why voltage has nothing to do with the amount. Current is how many electrons cross each second. Voltage is how hard each one is pushed. The amount of product depends only on how many crossed. So only the current and the time enter this sum. Voltage decides the energy bill, which is a different question.

Why this is a maximum and not a prediction. The sum assumes every electron did the intended reaction. Some never do. They drive a side reaction, or make hydrogen that recombines before it is collected. So the real mass is always lower. Comparing this number with what the separator actually caught is how a cell's current efficiency gets measured.

TAKES AS READ

1 mole of electrons carries 96,485 coulombs · splitting water to hydrogen takes two electrons per molecule · stoichiometry as a feed rate: turning a production target into kilograms in — taken as read

VERDICT 86W

Faraday's law is charge bookkeeping. Current times time gives the total charge. Divide by the charge one mole of product needs and you have moles. Multiply by the molar mass and you have grams. Voltage does not enter it. Voltage sets the energy cost, not the amount. And the answer is a ceiling rather than a forecast. It assumes every electron did the intended reaction, and some never do. Comparing this figure

with the hydrogen actually collected is how a real cell's current efficiency gets measured.

TAKEAWAY 16W

Charge and moles are one measurement in two units, once the electrons per molecule are known.

8.2 Four lines off the assay bench

REACTOR HALL • A PERSON • PROTOCOL

SCENE 31W

Stern's assay of the product stream has four lines on it. Herrera wants each one matched to what it implies about the loop before anybody argues about a set point again.

GUIDE 59W

Stern's assay has four lines, and Herrera wants each one placed before anybody argues about a set point again. Take them one at a time. Ask which part of the plant each line is reporting on. The bed? The recycle loop? The drier? And ask which line tells you least, because on an assay the biggest number usually does.

ASK 11W

Match each assay line to what it says about the loop.

BEHIND THE BUTTON 183W

Why the biggest number tells you least. A stream that is 96.4 per cent methane is a stream doing its job. Every plant's headline figure looks like that when nothing is badly wrong. The arguments live in the few per cent left over, because that is where anything unexpected has to be. Read the small lines first.

Why carbon monoxide is a reaction and not a contamination. Nothing out here delivers carbon monoxide into a sealed loop. It has to be made, and the only thing in the loop that can make it is the bed. So its presence says a second reaction is running on the same catalyst, competing for the same carbon. That is a selectivity problem, not a cleanliness one.

Why one line can mean two things. 400 parts per million of water is nothing at the reactor, which is making water as a product anyway. Two hundred metres downstream the pipe passes 273 K. There, the same 400 ppm is ice on an exchanger wall. An assay line only means something once you ask where the stream goes next.

TAKES AS READ

the loop returns unreacted gas to the reactor inlet • the product stream is dried before it is cooled • stoichiometry as a feed rate: turning a production target into kilograms in — taken as read

VERDICT 97W

Every line in an assay belongs to a part of the plant. The methane figure is the headline and tells you least. It only says the plant is doing its main job. Carbon monoxide did not blow in from anywhere. It is evidence of a second reaction on the same bed, taking carbon the methane wanted. Unreacted hydrogen is expected in a loop that recycles, and would be alarming in one that did not. Water past the drier is a maintenance item here. Two hundred metres downstream, at 273 K in the exchanger, it is an emergency.

TAKEAWAY 16W

An assay is a statement about the process, not only about the bottle it came from.

8.3 The bed that warns only at the outlet

WATER PLANT · A MODULE · CHOICE

SCENE 42W

The polishing column is a packed bed of ion-exchange resin with a fixed number of sites in it. Abiola has been asked how the plant would know the column was full, and he says the honest answer is that it would not.

GUIDE 68W

Abiola's honest answer is worth sitting with: the plant would not know. The column is a packed bed with a fixed number of exchange sites in it. Water goes in one end and out the other. Picture what happens to those sites as sol after sol of water passes through. Which ones fill first? Then ask of each option whether it fits a bed that fills that way.

ASK 15W

What happens at the outlet as the exchange front reaches the end of the bed?

BEHIND THE BUTTON 180W

What ion exchange is doing. The resin is covered in sites, and each site holds a harmless ion. It will swap that ion for one you want gone — chloride, here. Each site does this once. There is no regenerating it in service and no partial credit. A site is either free or taken.

Why the bed fills from the inlet and not evenly. The first water through meets the first centimetre of resin and gives up its chloride there. Only when those sites are taken does the next water travel further to find a free one. So the used region grows from the inlet downstream, as a front, with clean resin still waiting beyond it.

Why capacity is not a percentage, and why that is the trap. A filter that removes 95 per cent removes 95 per cent all its life. A capacity bed removes nearly everything, and then removes nearly nothing. There is no gradual warning between those two states. That is why Abiola wants an alarm on the outlet rather than a maintenance date on a calendar.

TAKES AS READ

an ion-exchange bed swaps unwanted ions for harmless ones · a packed bed holds a finite number of ions · rate constants, and how steeply they climb with temperature — taken as read · stoichiometry as a feed rate: turning a production target into kilograms in — taken as read

VERDICT 103W

An ion-exchange bed holds a fixed number of sites, and they fill from the inlet end. So a loading front creeps down the resin, with fresh sites ahead of it and full ones behind. While there is fresh resin left, the outlet stays clean. Then the front reaches the end and the outlet climbs fast toward the inlet concentration. That climb is called breakthrough. It is not instant, and its exact shape depends on flow, but it is a cliff rather than a slope. So the only warning is at the outlet — and only if somebody reads it often enough to act.

TAKEAWAY 15W

A bed with a capacity has two states, and there is no announcement between them.

8.4 What has to be paid for — Review

PLANT CONTROL · A MODULE · CALLBACK · CASEBOOK

SCENE 31W

Demir has four of the station's processes on the board. She wants them sorted before the quarterly energy plan is written. Two of them run downhill. Two have to be pushed.

GUIDE 66W

Two questions of each process, in this order. First: does it run downhill on its own? Or does something have to push it? That question decides whether there is a bill at all. Second, take the ones that do need pushing. Is the bill chemical or physical? And watch out for one process here. It runs downhill so willingly that stopping it is what costs money.

ASK 10W

Match each process to what it needs from the station.

BEHIND THE BUTTON 178W

Why direction and speed are separate questions. A negative free-energy change says a process can run on its own. It says nothing whatever about when. Making methane is downhill. It would still take years without a catalyst. So "it runs on its own" and "it runs" are two different claims. Only the first one is about thermodynamics.

Why the same reaction appears twice on this board. Splitting water and making methane are close to one reaction. It is read in two directions. One is uphill and has to be paid for electrically. The other is downhill and pays out. That is one number with two signs. So this station can run one step for free and never the other.

Why stopping something downhill costs money. Boiling is downhill. Heat arrives and liquid becomes gas without anyone helping. So keeping oxygen liquid means fighting a process that wants to happen. Every second, forever. That is what the refrigerator is for. A downhill process is not always the cheap one. It is only cheap if you wanted it to happen.

TAKES AS READ

a negative free-energy change means a process can run on its own · running on its own says nothing about how fast a process goes

VERDICT 89W

Direction and speed are separate questions. Only direction says whether there is a bill. Making methane runs downhill, and it runs slowly. That is exactly what a catalyst is for. Splitting water is close to the same reaction backwards. So it runs uphill, and the bill is the free-energy difference paid in electricity. Boiling runs downhill too. That is why keeping oxygen liquid costs work, every second, to stop something that wants to happen. Compression is uphill, and physical rather than chemical. It is paid for at the compressor.

TAKEAWAY 19W

Direction is what sorts a station's processes. It says which of them have a bill nobody can design away.

END OF THE SOL 14W

A measurement nobody asks for is a measurement that arrives later at full size.

The poison in the spent charge

PLAN CARD 68W

Sol 299. The dead charge came back from the assay bench an hour ago. Nickel particle size is normal, carbon deposit is only trace, but chlorine is concentrated at the inlet surface. That is the first finding able to distinguish the two explanations the plant has argued about for days. Today you identify the mechanism and then use the water-plant record to find when the contamination path opened.

BRIEFING 20W

The spent catalyst finally supplies evidence that can separate heat damage from poisoning and date when the real failure began.

WHAT YOU DECIDE 13W

Find out what has been taking the bed apart, and for how long.

9.1 One line off the spent charge

CATALYST BAY · A MODULE · CHOICE

SCENE 35W

The spent charge was assayed in three sections: the first 20 millimetres, the middle, and the outlet end. Kaur has the five findings on the board and is no longer sure of her own explanation.

GUIDE 83W

Kaur has stopped trusting her own explanation, which is the right moment to look at the material rather than the plant. Each of the four candidates would leave its own mark on the catalyst itself. Heat damage shows up in the size of the metal particles. Coking shows up as carbon on the surface. A bad batch shows up everywhere at once. So take each one and ask what the assay ought to show if it were true. Then read the five findings.

ASK 5W

Which explanation fits every finding?

BEHIND THE BUTTON 194W

How a catalyst dies of heat, and what it leaves behind. The nickel works as tiny particles, because small particles give a great deal of surface for their mass. Run them hot enough for long enough and neighbouring particles merge into bigger ones. That is sintering, and it destroys surface without destroying any nickel. It is permanent, and it is visible: the particles are measurably larger.

How poisoning can move down a bed. A poison arriving in the feed encounters the inlet first. If it is strongly captured there, the front loses activity before the back. As those sites fill, contamination penetrates farther downstream. A chlorine-rich inlet surface with a clean outlet is therefore evidence for a feed-borne chlorine-containing poison, not proof that dissolved chloride ions themselves travelled unchanged through every upstream process.

Why two causes with one symptom are settled in the material. Sintering and poisoning both look identical from the control room. Conversion falls; the hot spot moves. Neither reading tells them apart. The catalyst itself does. Particle size answers the first question, surface chlorine answers the second, and a sealed can sat on a shelf for eleven sols holding both answers.

TAKES AS READ

a catalyst can lose activity by heat or by a substance binding to it • heat makes small metal particles merge into larger ones

VERDICT 91W

One material observation settles the competing explanations. Nickel particle size matches fresh material, so the assay does not support sintering. Carbon deposit is only trace. A bad charge would be weak along its whole length, but the outlet end remains active. What remains is 1.9 percent chlorine on the inlet surface and none at the outlet: a spatial fingerprint of chlorine-bearing contamination arriving with the

feed and being captured where it first meets nickel. That also explains the walking hot spot—the reaction begins farther downstream as the inlet loses usable surface.

TAKEAWAY 17W

Two causes with one symptom are told apart in the material rather than in the plant readings.

9.2 The trace nobody read

WATER PLANT · A PERSON · CHOICE

SCENE 35W

The conductivity of the water leaving the polishing column is logged every sol. The trace runs flat for 260 sols, steps up over about 4, and runs flat and high from there to this morning.

GUIDE 70W

One trace, printed every sol, and nobody had a reason to read it until now. The shape is the evidence, so read the shape. Flat for 260 sols. Then four sols of rising. Then flat and high ever since. For each option, ask two things. What would the inlet record have to show if that were true? And would it give you a rise over four sols, or something else?

ASK 8W

What does the shape of that trace say?

BEHIND THE BUTTON 195W

What conductivity is measuring. Pure water barely conducts electricity at all. Dissolved ions carry charge, so the more ions in the water, the better it conducts. That makes a conductivity meter a cheap, continuous count of how salty the water is. It cannot say which ion. It can say how many, every sol, for years.

What a breakthrough curve looks like, and why four sols is the tell. A fresh exchange bed keeps the outlet clean while it has free sites left. As the loading front reaches the outlet, the reading climbs toward the inlet value and then stays there. Low, rise, high. The rise takes as long as the front takes to clear the last of the resin — sols, not hours.

Why the shape rules the other three out. A saltier feed would show in the inlet record too, and the inlet is flat across the whole period. A sensor reset happens in one sol, not four. A leak around a seal would have raised the trace from the day it was fitted, sloping, with no flat run before it. Only one candidate makes a flat line, a four-sol ramp, and a plateau.

TAKES AS READ

conductivity rises with the ions dissolved in water · an exchange bed holds a fixed number of ions · stoichiometry as a feed rate: turning a production target into kilograms in — taken as read

VERDICT 112W

The shape of the record carries the history. A fresh exchange bed keeps the outlet clean while it still has free sites. As the loading front reaches the outlet, the reading climbs toward the inlet value and then stays there. Low, then rising, then high — that is the breakthrough signature. A saltier feed would show at the inlet too, and the inlet is flat. A sensor reset moves a trace within one sol, and this took four. So the trace dates breakthrough to about sol 261, and says the plant has been sending

unpolished water into the electrolysis train ever since, opening a pathway for halide contamination to reach downstream feed.

TAKEAWAY 17W

A step in a record is a date, and a date turns a fault into a history.

9.3 Change the time, not the ceiling

REACTOR HALL • A MODULE • CONTROL

SCENE 33W

Herrera and Sundqvist have argued for a fortnight about whether the hot reactor is at equilibrium or simply short of reaction time. Stern gives them one controlled run before the production schedule resumes.

GUIDE 86W

The number you are watching is conversion — how much of the gas that went in came out as product. Two explanations are on the table. Either the gas leaves before the reaction has finished, or it has hit a ceiling the equilibrium sets. Three settings can be changed and only one of them tells those apart. Think about which. A response only counts if it beats the assay's own scatter. So change one setting, run it, put it back, and name the one conversion follows.

ASK 18W

Which controlled change separates a kinetic limit from an equilibrium limit, and does conversion follow it on reversal?

BEHIND THE BUTTON 188W

What the argument is actually about. A reaction that has not finished and a reaction that has hit its limit both give a low conversion. On one measurement they read exactly the same. The difference is what they answer to. More time helps the first and does nothing at all for the second. So the test has to be a change in time.

What each of the three settings does. Gas flow sets how long the gas spends over the bed. It changes the time available and leaves the ceiling where it is. Temperature and pressure both move the ceiling itself. That is why only one of the three can separate the two explanations — the other two change both things at once and settle nothing.

Why putting the setting back matters. Plants drift. A bed ages, a heater wanders, a feed composition shifts, and conversion moves a little while you work. So a rise after your change might not be your change. Restore the setting and watch conversion come back to where it was. That is how you tell your own effect from the plant's own drift.

TAKES AS READ

equilibrium is a ceiling that depends on temperature and pressure • conversion also depends on how long the gas spends in the bed • rate constants, and how steeply they climb with temperature — taken as read • stoichiometry as a feed rate: turning a production target into kilograms in — taken as read

VERDICT 98W

At fixed temperature and pressure, the ceiling does not care how fast gas crosses the bed. The rate does. So halving the flow doubles the contact time and leaves the ceiling exactly where it was. In this run conversion rose from 64 to 69 per cent, well past the assay's own scatter. It came back to 64 when the flow was restored. That means the gas had been leaving before the mixture reached its ceiling. The shortfall

was time, not thermodynamics. Temperature or pressure could not have answered it, because each moves the ceiling and the rate together.

TAKEAWAY 18W

Changing residence time at fixed temperature and pressure changes kinetic approach to equilibrium without moving the equilibrium ceiling.

9.4 Reading a blockage from outside the pipe — Review

COLD END · A MODULE · CALLBACK · DIAGNOSIS

SCENE 34W

A line on the oxygen side has slowed over two sols. Cho has four readings from around it, and the plant needs to know whether to warm the line out or open it up.

GUIDE 73W

Same discipline as the methane line, on a different pipe. Three of these five readings are doing the ruling out, so start with them rather than with the alarm. And look at the shape of the flow again. Something that closes in a minute is not the same story as something that closes over two sols. Then one extra question, because Cho has to choose an action: what is the blockage made of?

ASK 5W

Which explanation fits every reading?

BEHIND THE BUTTON 122W

Why a normal reading is a result. The alarm often fits several explanations; the quiet readings are what rule them out. A cold head at its set point says refrigeration is available, and a steady jacket pressure argues against insulation failure.

Why trace shape is evidence. A commanded valve change happens quickly and then stops. A solid layer accumulating on a wall can reduce flow gradually over many hours or sols.

Why material decides the repair. At this line's measured pressure and composition, the calculated carbon-dioxide frost/sublimation window is reachable by a controlled warm-up. Water ice would require a much warmer intervention. The procedure therefore uses the line's actual phase diagram rather than treating 195 K or 273 K as universal switches.

TAKES AS READ

phase boundaries are evaluated at the line's measured pressure and composition · the decline over two sols is inconsistent with a single instantaneous valve movement

VERDICT 82W

The diagnosis uses both trace shape and process conditions. A gradual two-sol loss of flow points to material accumulating in the line rather than a one-time valve command. The cold head and jacket readings rule out other causes. Once a solid blockage is the leading explanation, its identity matters because each substance has a pressure-dependent phase boundary. Use the line's measured pressure and composition to decide whether a safe controlled warm-up can clear it or whether the hardware needs a different intervention.

TAKEAWAY 19W

The same pressure-flow symptom can require a different repair depending on which material's phase boundary the line actually crossed.

The plant was arguing about temperature while a contamination path was taking the catalyst away.

The last spare charge

PLAN CARD 56W

Sol 300. The final spare catalyst charge is out of stores. If it is damaged, there is no replacement before launch. Today you bring it into service in a reversible order, spend the one hardware improvement the station can still make, and write the operating comparison that will tell later crews what this charge actually proved.

BRIEFING 20W

There is one catalyst charge left on Mars, so commissioning it has to produce evidence rather than merely restore production.

WHAT YOU DECIDE 13W

Get a bed back into service and decide what stops this happening again.

10.1 Bringing a charge into service

CATALYST BAY · A MODULE · SEQUENCE

SCENE 38W

The fresh charge arrives as nickel oxide on alumina, which is not a catalyst yet. Kaur has four operations written out and wants them in an order she can hand to a technician at three in the morning.

GUIDE 65W

What arrives in the can is nickel oxide. Nickel oxide is not a catalyst. All four operations will happen, so the question is what each one makes possible, or safe. Two things to hold on to. Hydrogen and oxygen together at 350 °C is not something you want inside a reactor. And nobody hands full production feed to a charge whose activity is still unknown.

ASK 10W

Order the four operations for bringing the charge into service.

BEHIND THE BUTTON 200W

Why the charge has to be reduced before it works. Nickel does the work here, and what ships in the can is nickel oxide. That is nickel with oxygen attached. Pass hot hydrogen over it and the oxygen leaves as water, and nickel metal is left behind. This is reduction, the same word as ever: something gained electrons. Until it happens there is no catalyst in the reactor. Only its ingredients.

Why the purge is not a formality. The reduction step needs hot hydrogen. If oxygen is still in the vessel when that hydrogen arrives, the two react with each other instead, and they do it fast. Nitrogen is used to push the oxygen out first because nitrogen reacts with nothing here. It is a step whose only purpose is to make the next step safe.

Why the low-flow check comes before the comparison. There is one spare charge on the planet. A tenth of production flow is enough to show whether the bed lights off at all. If it does not, almost nothing has been risked. Only once the bed is known to be active is the sol-270 comparison worth running. That run leaves the next crew a number.

TAKES AS READ

nickel oxide has to be reduced to nickel metal before it is active · hydrogen at temperature will reduce a metal oxide

VERDICT 89W

Each operation makes the next one possible, or safer. The nitrogen purge removes oxygen before hot hydrogen goes in, because those two react with each other given the chance. Reduction then turns nickel oxide into nickel metal. Until that is done there is no catalyst in the reactor. A light-off check at a tenth of flow shows whether the charge is active, at almost no risk. Only then is a matched comparison against the sol-270 run worth running. It leaves the next crew a number instead of an impression.

TAKEAWAY 18W

A part that cannot be replaced is proved at low rate before it is trusted at full rate.

10.2 One improvement, four candidates

ELECTROLYSIS HALL · A PERSON · SCIENCETANK

SCENE 37W

The station has parts and crew time for one improvement. One catalyst charge is already lost, and there is no spare after the new one. Achebe puts four candidates on the board before anyone starts cutting pipe.

GUIDE 102W

Here is what the last fortnight established, and it is worth having straight before you spend anything. The plant ran 38 sols on water no column was polishing. It lost a catalyst charge to that water. Conductivity was logged on every one of those sols, and read by nobody. The array has not been the limit on a single sol this season. There are already four probes in the bed. A second column in series would let the lead bed be swapped at breakthrough, and it needs pipe and a shutdown. And one catalyst charge is left, with no spare behind it.

ASK 12W

Fund the change that would most alter what the plant does next.

BEHIND THE BUTTON 188W

Why the whole spread is graded. Funding is not a vote for one idea. A portfolio says three things at once: what you think is likely, what is worth hedging against, and what is not worth doing at all. Most of the information is in the last two. Backing the right proposal while quietly funding a bad one is a worse answer than it looks. The small numbers count as much as the big one.

What the three numbers are for. Thirty-five is what makes a lead a lead. Below it you have hedged rather than chosen. Fifteen is the most that can sit on unsupported work before it stops being a rounding error. Past that it is a second opinion nobody argued for. Twenty is the floor under work you have already called strong, because funding it too thin to finish spends the money and buys nothing.

Why there is a floor on the total. Points held back are not caution. They are a decision not to decide, taken with somebody else's money and somebody else's deadline. The floor is what forces the panel to say something.

TAKES AS READ

the polishing column gives no warning as it fills · the plant has no spare catalyst charge after this one · stoichiometry as a feed rate: turning a production target into kilograms in — taken as read

VERDICT 99W

The plant did not lack a measurement. Conductivity was recorded every single sol. What it lacked was a rule that turned the measurement into an action. An alarm on the feed water would have made breakthrough visible while the catalyst could still be saved. A second column adds capacity, and it still needs something to tell you when to switch over. Extra array power addresses nothing that failed, and the bed

already has four probes. The change worth funding first is the one that closes the gap between a signal you already have and a loss you cannot undo.

TAKEAWAY 16W

The work worth doing first is the work that would have caught the fault nobody caught.

10.3 The tank pressure that is not a leak

COLD END · A MODULE · CHOICE

SCENE 34W

The methane tank has climbed about 0.3 bar a sol for 4 sols. The mass in the tank is unchanged across the same four sols, and Moreau is being asked whether to vent it.

GUIDE 73W

Moreau is being asked to vent propellant, so it is worth being sure first. Two numbers matter and they disagree with each other. Pressure is climbing. Mass is not moving. Start from what would have to be true for each option, and check it against the mass. Then keep one fact in reserve: a tank holding liquid with vapour above it sits at the vapour pressure of whatever temperature it happens to be.

ASK 13W

The pressure is rising and the mass is not falling. What is happening?

BEHIND THE BUTTON 196W

What vapour pressure is. Put a liquid in a sealed tank and some of it leaves as gas. That gas builds up until as many molecules are returning to the liquid as are leaving it. The pressure it settles at is the vapour pressure, and it depends on one thing only: the temperature. Warm the tank and more molecules have the energy to leave, so the pressure settles higher.

Why that makes a pressure gauge a thermometer. As long as there is liquid and vapour together in the tank, the pressure is pinned to the temperature. It is not a free variable. So a pressure reading climbing steadily for four sols is a temperature reading climbing steadily for four sols, told in different units.

Why the mass is the reading that decides it. Gas leaving a tank — through a leak, or a vent — takes mass with it. There is no way around that. So a flat mass across four sols rules out anything that works by losing contents. So what is left has to raise the pressure while keeping every molecule inside the tank. Warming is the only candidate on the list that does.

TAKES AS READ

a liquid in a closed tank sits under the pressure of its own vapour · vapour pressure rises steeply with temperature

VERDICT 98W

A tank holding liquid with vapour above it sits at the vapour pressure of its own temperature. So while both are present, the pressure gauge is a thermometer. Here the mass has not moved for four sols. A leak or a sustained vent takes mass out, so the inventory rules both out. That leaves warming, and the pressure is simply following the temperature up. The right response is to go and find the heat — insulation, a valve conducting warmth in, a refrigerator falling behind. Venting propellant because a gauge rose spends the tank to treat a symptom.

TAKEAWAY 15W

Pressure in a cryogenic tank is a reading of temperature until the mass says otherwise.

10.4 Why the hotter pass gave less — Review

REACTOR HALL · A MODULE · CALLBACK · CHOICE

SCENE 33W

Herrera has the equilibrium constant for the methane assembly at four temperatures: 500 K, 550 K, 600 K and 650 K. It falls by more than a factor of 10 across that range.

GUIDE 66W

A table of the equilibrium constant against temperature, and it falls. Start with what the constant is a property of, because two of these options assume it is a property of something else. Then read the direction. A constant that falls as the bed gets hotter is telling you which side the heat is on. That is a fact about the reaction, not about the equipment.

ASK 11W

What does a constant falling steeply with temperature tell the plant?

BEHIND THE BUTTON 198W

What K is a property of, and what it is not. The equilibrium constant belongs to the reaction, and it answers to temperature and to nothing else. Not to pressure. Not to the catalyst. Not to how much of anything you put in. Two of the options on this board get that wrong, and both are tempting, because both name something that really does affect the plant.

How to read the sign of the heat off a table. If K falls as the temperature rises, the reaction is giving heat out. Heat behaves as one of its products, so supplying more pushes the balance back. If K rose instead, the reaction would be taking heat in. That is exothermic and endothermic read straight off a column of numbers, with no apparatus involved.

Why this table is the plant's whole design problem, written in advance. Yield wants a cold bed. Rate wants a hot one. A factor of ten in K across 150 degrees says how expensive that argument is. The operating temperature is picked where a good enough catalyst makes a good enough yield reachable in the time available. The table is what tells you where to look.

TAKES AS READ

the equilibrium constant depends on temperature and on nothing else · this reaction gives out heat

VERDICT 83W

The direction the constant moves in tells you the sign of the heat. Falling with temperature means heat behaves as one of the products, so supplying more of it pushes the balance back. That is an exothermic reaction read straight off a table. And the consequence is this plant's whole design problem. Yield wants a cold bed. Rate wants a hot one. So the operating temperature is chosen where a good enough catalyst makes a good enough yield reachable in the time available.

TAKEAWAY 14W

A table of constants against temperature is the plant's ceiling written out in advance.

END OF THE SOL 13W

A part with no replacement is commissioned by procedure rather than by enthusiasm.

Forty clean sols

PLAN CARD 69W

Sol 340. Forty sols after the last spare charge went in, conversion is holding inside its qualified operating window and the polishing outlet has stayed below its warning line. Nothing is on fire, which is exactly why today matters. You close the electrical ledger against what the array supplied, close one-pass carbon bookkeeping, and check that the tank is being filled without hiding losses in a cheerful production number.

BRIEFING 24W

The replacement charge has held for forty sols on qualified feed, giving the team a quiet checkpoint to reconcile energy, carbon and cryogenic losses.

WHAT YOU DECIDE 11W

Close the plant's own books while there is nothing to firefight.

11.1 The bill for a kilogram of hydrogen

PLANT CONTROL · A MODULE · BALLPARK

SCENE 30W

Demir is closing the sol's energy ledger. The stacks made 6.9 kg of hydrogen. He wants their electrical energy use written beside the array output before he signs the sheet.

GUIDE 72W

Demir wants what the stacks spent, not what the hydrogen stores. Those are different numbers. A kilogram of hydrogen carries about 39.4 kWh on the stated fuel-value basis, while this stack has measured electrical use near 55 kWh/kg. The gap includes the real cell's voltage losses, balance-of-plant loads and thermal management; it is an observed efficiency gap, not a claim that every joule above the reversible voltage becomes one simple heat term.

ASK 11W

Estimate the electrical energy the stacks spent making hydrogen this sol.

BEHIND THE BUTTON 181W

Why a kilogram of hydrogen has two energy numbers. One is what it stores: burn it, or run it through a fuel cell, and about 39.4 kWh comes back out. The other is what it cost to make: about 55 kWh through this stack. Both are correct. They answer different questions, and a ledger wants the second.

Where the gap comes from. A practical electrolyser operates away from the reversible limit and also powers pumps, controls and thermal management. Electrode kinetics and resistance dissipate energy, while the cell's full heat balance is referenced to the thermoneutral as well as reversible voltage. The plant therefore uses measured kWh per kilogram for its ledger instead of reconstructing the whole loss from one voltage subtraction.

Why using the wrong figure matters here. Put 39.4 into the sum and the sol's bill reads 272 kWh. The true figure is 380. That is a quarter of the entry missing from an energy ledger, on a station whose array is the only source of power. A cheerful number in the wrong column is worse than no number.

TAKES AS READ

electrical energy can be counted in kilowatt-hours · a stack costs more energy per kilogram than the hydrogen itself carries

VERDICT 75W

A kilogram of hydrogen has an energy content and this plant has a measured electrical cost to make one. Those numbers answer different questions. At 6.9 kg and 55 kWh/kg, the stack used about 380 kWh. The difference from the hydrogen's stored energy reflects real electrolysis inefficiency and balance-of-plant costs; it should not be simplified to 'everything above 1.23 V became heat.' For the station ledger, the measured 55 kWh/kg is the quantity that matters.

TAKEAWAY 25W

What a product costs to make and what it stores are two different numbers, and the gap is the price of running at a rate.

11.2 The space left at the top

COLD END · A PERSON · CHOICE

SCENE 36W

Anaya Bhatt, the ascent vehicle systems engineer, stops the methane tank at 95% by volume. Moreau has been asked why the remaining 5% is not simply more propellant, with the window as close as it is.

GUIDE 77W

The window is close and 5 per cent of a tank is a lot of propellant, so Moreau's answer had better be a good one. Think about what happens to that tank between filling and launch. Nobody draws from it. Heat still reaches it. So ask of each option whether it is really about the liquid, or about the gas above the liquid. That gas space has a name, by the way. It is called the ullage.

ASK 11W

Why is the flight tank filled to 95% rather than full?

BEHIND THE BUTTON 183W

Why a cryogenic tank always has gas in it. The liquid is sitting at its boiling point. Insulation slows heat down and never stops it. So some liquid is always turning to vapour, and that vapour has to occupy space. This is not a fault to be engineered away. It is the condition the tank permanently lives in.

What happens to a tank filled to the brim. Vapour still forms, and now there is nowhere for it to go. Gas that cannot expand raises the pressure instead, and it does so fast. The relief valve opens and starts throwing propellant overboard to protect the tank. A small heat leak has become a loss of fuel, and the vehicle has not burned a gram.

Why 5 per cent rather than 1 or 20. The ullage has to swallow every kilogram that boils off between filling and launch, without the pressure reaching the relief setting. Estimate the boil-off rate. Multiply by the wait. Add margin for a bad sol. The answer comes out at a few per cent. It is a calculation, not a habit.

TAKES AS READ

a liquid at its boiling point turns to vapour when heat reaches it · a closed vessel has a pressure it is rated to

VERDICT 104W

A cryogenic propellant tank is designed with vapour space and a pressure-management scheme because heat keeps arriving, because heat keeps arriving however good the insulation is. The gas space above the liquid is the ullage, and its job is to give that new vapour room. Fill the tank to the brim and the room is gone. Vapour then raises the pressure instead, quickly, and the relief valve has to vent propellant to protect the tank. So a small heat leak becomes a fuel loss before the vehicle has burned a gram. The 5 per cent is what buys the wait between filling and launch.

TAKEAWAY 14W

A cryogenic tank is designed around what happens to it between filling and use.

11.3 Close the carbon pass

REACTOR HALL · A MODULE · BALANCE

SCENE 35W

Herrera pins one pass of the carbon assay to the board: 100 moles of carbon dioxide entered. Methane and carbon monoxide are measured at the outlet, and the recycle meter records the unreacted return stream.

GUIDE 72W

A hundred moles of carbon went into the bed, so a hundred moles have to come out of it somewhere. Methane and carbon monoxide are measured at the outlet. The recycle meter records what came back unreacted. Read every stream, then count only the carbon that belongs to this pass. And keep one distinction in mind as you go: carbon that comes back is not the same as carbon that is gone.

ASK 21W

Where are all 100 carbon moles after 1 pass, and what does that imply for overall methane yield after repeated recycle?

BEHIND THE BUTTON 177W

Why one pass is not the yield. A single pass converts part of the feed. The rest comes back and goes round again. So the overall yield after many passes can be far higher than the conversion on any one of them. That holds as long as nothing is lost on each circuit — which is what makes the recycle number worth checking.

What counting a stream actually claims. Reading a meter is free. Counting the number says this carbon belongs to the same hundred moles that went in. Methane and carbon monoxide are both products. Unreacted carbon dioxide is not a product, but it is still carbon, and it still has to appear somewhere in the ledger.

Why a gap would change the plan. Carbon that leaves the loop each pass is carbon the recycle can never recover. It might be laid down on the catalyst, vented, or dissolved in water. A small loss on every circuit compounds into a hard ceiling on overall yield. That ceiling is the number the ascent vehicle's schedule rests on.

TAKES AS READ

the recycle loop returns unreacted gas to the inlet · every carbon atom entering the bed leaves it in something

VERDICT 94W

A one-pass conversion of 64% does not mean 36% of the feed is lost. The ledger closes as 64 moles of methane, 2 of carbon monoxide, and 34 of unreacted carbon dioxide going back to the inlet. That recycle term is inventory, not waste. It gets another chance next pass, and another after that. So if the selectivity holds, the split settles at the ratio of the two things that cannot come back: 64 to 2. That is about 97% methane and 3% carbon monoxide. Per-pass conversion is 64%. Overall conversion is nearly everything.

TAKEAWAY 18W

Per-pass conversion counts what reacts now; overall conversion counts what is eventually recovered after unreacted feed is recycled.

11.4 Getting water out of frozen ground — Review

WATER PLANT · A MODULE · CALLBACK · SEQUENCE

SCENE 37W

A load has come in at 3% water instead of eight, and the same four operations have to run on it. Sora Nakagawa, the drill and excavation operator, wants the order confirmed before the hopper is sealed.

GUIDE 70W

Same four operations, a much poorer load. So ask a different question this time: what would you be unable to tell apart if a step were missing? A lean load gives little water. So does a broken plant. Something on this list is what separates those two. And note that the warming has a ceiling as well as a floor, because perchlorate salts come apart when they are heated hard.

ASK 7W

Order the operations for a lean load.

BEHIND THE BUTTON 179W

Why a lean load needs the sample and a rich one does not. Expect eight per cent water and get eight, and nothing needs explaining. Expect nothing in particular and get very little, and you cannot tell a poor load from a plant fault. The sample turns the second sol into a measurement instead of a shrug. That is why it moves to the front when the ground gets worse.

Why the warming has a ceiling. Oxychlorine salts such as perchlorates can decompose when heated hard and create reactive chlorine-bearing products. The plant therefore heats only enough to recover water, and treats high-temperature decomposition as a separate contamination pathway—not as ordinary perchlorate simply dissolving into chloride.

Why the order is the same on a poor load as on a good one. A leaner load carries the same salts in less water. It does not change what depends on what: still sealed before warmed, still condensed before polished. What changes is the yield, and the value of having predicted it. A poorer load changes the numbers and not the sequence.

TAKES AS READ

warming ice-bearing soil releases water as vapour at low pressure · a leaner load carries the same salts in less water

VERDICT 75W

Sampling comes first on a lean load because without an expected water yield, poor ground and a plant fault look the same. Warming then has both a floor and a ceiling: enough heat to recover the water, but not so much that oxychlorine salts decompose and create new reactive chlorine-bearing contamination. The vapour is condensed and measured against the predicted yield, and the liquid is polished at the qualified flow before it is sent downstream.

TAKEAWAY 15W

A poorer load changes the yield of a process and not the order of it.

END OF THE SOL 14W

The sol to check the books is the sol when nothing is going wrong.

Dust over the array

PLAN CARD 65W

Sol 420. A regional dust storm crossed Arcadia Rise overnight. Measured array output is about one fifth of a clear sol, but the sun sensor and optical-depth model describe only the direct beam. Today you calculate what comes straight through the dust, keep that model separate from total photovoltaic output, and decide which loads can stop without destroying propellant or putting the habitat at risk.

BRIEFING 25W

A regional storm cuts total array output to about one fifth of a clear-sol value; the direct-beam model predicts something different and the distinction matters.

WHAT YOU DECIDE 10W

Decide what the station stops doing when the power halves.

12.1 What the dust leaves of the sun

ELECTROLYSIS HALL · A MODULE · BALLPARK

SCENE 45W

Petrov has two different quantities on his panel. The optical-depth instrument reports $\tau = 2.4$, while the array meter says total electrical output is about one fifth of a clear-sol value. He wants the direct beam calculated without pretending it is the whole photovoltaic measurement.

GUIDE 44W

Use $I = I_0 e^{-\tau}$ only for the direct solar beam. The array also receives diffuse light scattered by dust and its electrical output includes panel geometry, temperature and conversion efficiency. Calculate the direct-beam scale, then keep it separate from the measured one-fifth total output.

ASK 10W

Estimate the direct beam reaching a panel through this dust.

BEHIND THE BUTTON 181W

Why attenuation multiplies rather than subtracts. Picture the dust as a stack of thin layers. Each layer stops the same fraction of whatever light reaches it — not the same amount. So the first layer takes a lot and the tenth takes a little, because there is less left for it to take. Repeated fractions of fractions is what an exponential is.

What optical depth means as a number. Depth 1 leaves about a third of the beam. Depth 2 leaves about a tenth. Depth 2.4 leaves about a twelfth. So going from a clear sol at 0.4 to a storm at 2.4 is not five times worse. It is roughly sevenfold, because the two extra units of depth each divide again.

Why the direct beam is not the array's output. What $e^{-\tau}$ describes is light that came straight through untouched. Dust also scatters light sideways, and some of that still lands on a panel from the sky. So the array does better than the direct beam alone suggests. It does not do anything like as well as a clear sol.

TAKES AS READ

each unit of optical depth cuts a beam by a further factor • a beam is attenuated as an exponential of the depth it passes through • stoichiometry as a feed rate: turning a production target into kilograms in — taken as read

VERDICT 83W

The Beer–Lambert-style attenuation model gives the direct beam: $490 \times e^{-2.4} \approx 44 \text{ W/m}^2$, about nine percent of the above-dust beam. The array meter, however, reports total electrical output near twenty percent of a clear sol. Those do not conflict automatically. Dust also scatters light into the sky, panels collect some diffuse illumination, and electrical output depends on more than the straight-through beam. The model and the measurement constrain different quantities, which is exactly why they should not be substituted for one another.

TAKEAWAY 19W

Optical depth predicts direct-beam attenuation; total photovoltaic output is a separate measurement with additional light paths and hardware response.

12.2 What the station stops doing

PLANT CONTROL · A MODULE · ALLOCATE

SCENE 40W

The dust storm has cut the sol from about 2,000 kilowatt-hours to 600. Tank refrigeration is already drawing power, and six other demands are queued. Marion Wells, who commands the station, wants the sacrifices written down before breakers are opened.

GUIDE 78W

Six hundred kilowatt-hours for the whole sol, against about two thousand on a clear one. Watch the answers list rather than the equipment. It shows what the station can still say at the end of the sol under your plan. The habitat is not a load and comes out first. After that, rank what is left by one question: if you stop this, can you get it back? Some of these you can. One of them you cannot.

ASK 17W

600 kilowatt-hours. Build the sol's plan, and know what it stops the station being able to say.

BEHIND THE BUTTON 189W

Why reversibility is the ranking, not power drawn. Digging can wait — the ground will still be there. Batteries can recharge on the next clear sol. Electrolysis can be made up later if there is schedule margin left. None of those losses is permanent. Refrigeration is different in kind, and that is what puts it at the top rather than its 210 kWh.

Why stopping the refrigerator costs more power than it saves. Propellant already in the tanks took sols of array output to make. Stop cooling it and heat keeps arriving, so it boils away. The only way to get it back is to make it again, from the beginning, with power you do not have during a storm. Saving 210 kWh today to spend a thousand next week is not a saving.

Why writing the sacrifices down matters. Breakers opened without a record become an argument afterwards about what was agreed. Demir and Wells want the list because the storm will outlast everybody's memory of the reasoning. A plan that names what it gave up can be defended. A plan that only names what it kept cannot.

TAKES AS READ

stopping a process can be reversible or costly to reverse · a cryogenic tank warms up when its refrigerator stops

VERDICT 92W

Rank the loads by what stopping them costs to undo, not by what they draw. Digging can wait; the ground stays where it is. Batteries recharge on the next clear sol. Electrolysis can be made up later if there is margin. Refrigeration cannot. Stop it and heat keeps entering the tanks, and propellant that took sols of array output boils away for good. Habitat power is protected separately, because the crew is not a load.

So the scarce pool protects what cannot be remade, then buys whatever single job is left affordable.

TAKEAWAY 16W

Loads are ranked by what stopping them costs to undo, rather than by what they draw.

12.3 Running the plant with no digging

WATER PLANT · A PERSON · CHOICE

SCENE 35W

The drill is off and the hopper is empty. The holding tank still contains 1,100 litres, while the reactor draws about 60 litres per sol. Abiola is asked whether the whole water plant must stop.

GUIDE 73W

The drill has stopped and Abiola is being asked whether everything downstream stops with it. Do the arithmetic before you read the options. Eleven hundred litres in the tank, sixty litres a sol going to the reactor. How many sols is that? Then ask of each option whether it stores water or makes it, because those are not the same kind of help. And remember what the Sabatier reaction puts out besides methane.

ASK 13W

What lets the reactor keep running through a storm with the drill stopped?

BEHIND THE BUTTON 205W

What a buffer does for a plant. Steps in a chain run at their own rates and break at their own times. A tank between two of them lets the upstream one stop without the downstream one noticing. Eleven hundred litres at sixty a sol is about eighteen sols of cover, and no storm recorded at this site has lasted that long. The tank is not spare capacity. It is the reason a stopped drill is not a stopped plant.

Why the reaction gives some water back. $\text{CO}_2 + 4\text{H}_2 \rightarrow \text{CH}_4 + 2\text{H}_2\text{O}$. Two moles of water come out for every mole of methane. The cold end condenses that water on its way down. Returned to the loop, it is water the drill did not have to cut. The plant is therefore not simply consuming water. It is recycling part of its own output.

Why the other two candidates cannot help. A polishing column holds ions on a resin, not litres in a vessel — there is no useful volume in it. And a condenser pulls water out of a stream that already has water in it. The atmosphere here is six millibars and nearly dry, so there is nothing out there to condense.

TAKES AS READ

the reactor's water demand is met from a holding tank rather than from the drill directly · the Sabatier reaction makes water as well as methane

VERDICT 99W

Two buffers stand between the reactor and the drill. The first is stored water. 1,100 litres at 60 a sol is about 18 sols of cover, and that is longer than any storm recorded here. The second is the reaction itself: $\text{CO}_2 + 4\text{H}_2 \rightarrow \text{CH}_4 + 2\text{H}_2\text{O}$ makes two moles of water for every mole of methane. The cold end condenses part of that back into the loop. So digging can pause without the reactor stopping. Which is the wider lesson — a plant survives a broken machine on the inventories between its steps, not on the machines.

TAKEAWAY 17W

A buffer between two steps is what lets one of them stop without the other one noticing.

END OF THE SOL 20W

A model of the direct beam and a measurement of total array output are related evidence, not the same quantity.

Three failures after the storm

BEFORE THE SOL — HUNT

Six regolith bins, and the tally says four

The feedstock tally is what the whole propellant schedule is planned against, and two bins are not where it says they are. A bin nobody can find is either full and idle or empty and counted, and those two mean very different launch dates.

PLAN CARD 60W

Sol 421. The sky cleared, but the storm left three problems: oxygen boil-off is high, the battery bank is at 30 percent, and the spare catalyst still needs its comparison run. All three matter. Today you follow the radiator fault to the oxygen vent, make the catalyst comparison fair, and decide which problem spends something unrecoverable every hour it waits.

BRIEFING 22W

The storm leaves three urgent problems, and the plant must rank them by irreversible loss rather than by which alarm sounds loudest.

WHAT YOU DECIDE 13W

Rank three faults that arrived together, and act on them in that order.

13.1 Follow the heat to the sky

COLD END · A MODULE · CHAIN

SCENE 35W

Oxygen boil-off has jumped from 12 to 26 kilograms a sol since the storm. Moreau opens the refrigeration schematic beside the tank. Jacket pressure is normal, while the compressor is already at its current limit.

GUIDE 74W

Boil-off has doubled and nothing has broken, which means something in the heat path has changed. So follow the heat. It starts in the liquid oxygen and has to end up somewhere outside the station. Every step between is a handover from one thing to the next. Build that path in order first. Then ask which single step is now holding the whole chain up. Two readings are evidence about that, so use them.

ASK 19W

Trace the heat from the liquid oxygen to its final sink. Which required transfer is now governing the loss?

BEHIND THE BUTTON 214W

Why boil-off is a heat measurement in disguise. Liquid oxygen sits at its boiling point, so every joule reaching it turns some into gas. Nothing warms up. Something evaporates instead. So 12 kilograms a sol becoming 26 says heat is arriving about twice as fast as it was. The tank has not changed. What reaches it has.

What a refrigerator actually does, and where the heat ends up. It does not destroy heat; nothing does. It picks heat up at a cold place and puts it down at a warm one. The compressor is what makes the refrigerant hot enough to give it up. But it still has to be given up to something. On a planet with no water and thin air, the only thing left is the sky. A radiator plate is how you talk to it.

Why a chain is only as good as one link. Heat has to make every handover to get out. Fail one and the whole path backs up, however well the others are working. So a compressor already at its limit is not the cause of the extra heat. It is the reason none of the extra can leave. And dust on a radiator breaks nothing. It only makes the last handover worse, which is enough.

TAKES AS READ

a refrigerator has to reject its heat somewhere · dust settles on any surface left out in a storm

VERDICT 102W

A refrigerator does not destroy heat. It moves it. Heat leaves the tank into the refrigerant. The compressor raises the refrigerant hot enough to be rid of it. The condenser hands it to the radiator, and the radiator gives it to the sky. Every step is a handover, and the chain is only as good as its worst one. The vacuum jacket is a different path — heat leaking in — and its normal pressure rules that out here. Dust has weakened the last handover of all. So the cold end warms while the compressor runs flat out, and nothing has failed.

TAKEAWAY 20W

A refrigerator can lose capacity because heat cannot leave its warm end, even when the compressor and insulation are intact.

13.2 Comparing two charges honestly

CATALYST BAY • A MODULE • CHOICE

SCENE 36W

Kaur wants the new charge compared against the sol-270 run before it is trusted with the rest of the season. The plant is at a different temperature, flow and pressure than it was on sol 270.

GUIDE 66W

Kaur wants to know about the charge, and only about the charge. But conversion answers to four things at once: temperature, pressure, flow, and the catalyst itself. Three of those are settings you choose. One is the thing being tested. So ask of each option whether it removes the other explanations, or quietly adds another one. A comparison is only about the variable you deliberately changed.

ASK 14W

What has to be true for a comparison between two charges to mean anything?

BEHIND THE BUTTON 180W

What a controlled comparison is for. Conversion has four inputs here, and you only want to learn about one of them. So hold the other three still. Then any difference you measure has nowhere else to have come from. This is what people mean by controlling a variable, and it is the entire reason the sol-270 settings are worth digging out of a log.

Why best-for-each is the tempting wrong answer. It sounds generous and it sounds rigorous. But suppose charge A runs at 580 K and charge B at 600 K. The two conversion figures now differ by the charge and by 20 degrees, with no way to separate them afterwards. Two numbers measured under different conditions cannot be subtracted. They are answers to two different questions.

Why more data does not repair an unmatched comparison. A whole sol of each charge gives you a much more precise measurement of the wrong quantity. Scatter shrinks; the confusion does not. That is worth remembering generally: precision and validity are separate properties, and only one of them improves by running longer.

TAKES AS READ

conversion depends on temperature, pressure, flow and the catalyst together • a comparison is only about the variable that was deliberately changed • stoichiometry as a feed rate: turning a production target into kilograms in — taken as read

VERDICT 90W

Conversion answers to four things at once. Temperature, pressure and flow are settings you choose. The catalyst is the thing being tested. Let the settings differ between two runs and the comparison mixes the charge in with the conditions, with no way to pull them apart afterwards. Match the settings to sol 270 and those other explanations are gone, so what is left can fairly be credited to the charge. Running each at its own best condition sounds fairer and is not. It produces two numbers that cannot be subtracted.

TAKEAWAY 10W

A comparison is worth what its matched conditions are worth.

13.3 Which of the three cannot wait

PLANT CONTROL • A PERSON • CHOICE

SCENE 32W

The three problems are on the board with the power to fix one of them today. Wells wants the order and the grounds for it in the log before anybody starts work.

GUIDE 77W

Three people want to be first and all three have a case, so the ranking has to come from something other than who is asking. Use the same test you used on the storm plan. For each job, ask what the delay actually costs. Some delays cost time, and time can be made up. One of them is spending something the station has already made, every sol it waits. That is not the same kind of cost.

ASK 8W

Which of the three is dealt with first?

BEHIND THE BUTTON 198W

The difference between a delay and a loss. Some jobs sit in a queue and cost nothing while they wait. The battery bank refills on the next clear sol. The comparison run is information, and information keeps. Other jobs are spending something every sol they are not done. Fourteen extra kilograms of oxygen a sol is propellant that took array output to make, and it is gone. Rank by that, not by urgency of tone.

Why boil-off is the one that cannot wait. The oxygen in that tank is stored work. Sols of power went into splitting the water and cooling the product. Let it boil and there is no recovering it. You have to start again from atmosphere and ice. Every other job on this board can be picked up on the next clear sol for exactly what it would have cost on 421.

Why finishing something is not a reason to choose it. Picking the job that fits in a sol ranks by convenience. It feels like progress because something gets closed, and it says nothing about what the other two cost while they wait. A ranking has to be about consequences. Neatness is not one.

TAKES AS READ

propellant vented from a tank cannot be recovered • a comparison run and a battery charge can both be done later

VERDICT 82W

Rank by what the delay costs, and whether it can be got back. Boil-off is the loss of something already made. Fourteen extra kilograms a sol, and no later work returns them. Batteries recharge on the next clear sol at no permanent cost. A comparison run is information, and information waits for nothing so long as the bed is watched. Only one of the three is spending an asset while it sits in the queue. That is the one that goes first.

TAKEAWAY 9W

What cannot be undone goes first, whatever is loudest.

13.4 What the plant is protecting now

PLANT CONTROL · A MODULE · SPOT

SCENE 40W

Loads come up on the board through the sol. The station runs to one power rule at a time. Demir changes it as the sol does: after the dust thickens, after the tank pressure rises, after the window gets closer.

GUIDE 67W

Keep the loads the current rule protects, and let the rest drop. One thing to know before you start. The rule at the top of the board changes during the sol, and nobody announces it. What gets scored is the loads either side of a change. Those are the only ones that show whether you are reading the rule, or working to the last one you saw.

ASK 11W

Protect what the current rule protects, and keep watching the rule.

BEHIND THE BUTTON 200W

Why the rule changes at all. On a clear sol the plant protects production, because the tank has a schedule to meet. In a storm it protects the crew and the propellant already made, because neither can be remade. Close to the window it protects whatever the launch depends on. Three situations, three lists, and the same eight loads on the board.

Why a load can be right under one rule and wrong under the next. The stacks are production, so they are protected on a clear sol and dropped in a storm. Tank pressurisation counts twice. It serves the launch, and it protects what cannot be remade, so it survives two rules of the three. Nothing on this board is simply important or unimportant. It depends what is being protected.

What the changeover actually costs. Picture a desk still running the production rule through the first hour of a storm. It keeps the stacks warm and lets the liquefier go off. The propellant that boils away took forty sols of array output to make, and it has to be made again from the beginning. The instruction was withdrawn; the cost is paid by whoever was still working to it.

TAKES AS READ

a station with limited power keeps some loads and drops others

VERDICT 130W

Three rules run across the sol, and each protects a different part of the board. Production first. Then whatever cannot be remade. Then whatever the launch depends on. A load can answer two rules at once, which is what makes a switch cost something real. So the panel scores the window either side of each change. A load protected by no rule at all is dropped correctly by somebody who has read nothing. It proves nothing. The station's own version of this is the first hour of a dust storm. The rule has moved to protecting what cannot be remade. A desk still running the

production rule keeps the stacks warm and lets the liquefier go off. That boils away propellant which took forty sols of array output to make.

TAKEAWAY 15W

The cost of a withdrawn instruction is paid by whoever is still working to it.

END OF THE SOL 15W

Three reasonable people wanting to be first is a ranking problem rather than an argument.

The batch you cannot take back

PLAN CARD 72W

Sol 445. A 410-kilogram methane batch is in the transfer tank and one assay line is outside the vehicle specification. There are thirty sols until the planned full-tank date—just enough to remake roughly this much at the plant's present rate. Once the loading valve opens, the batch cannot be taken back out. Today you decide whether it flies, then reconcile the chemistry's 4:1 combustion ratio with the vehicle's deliberately fuel-rich loading specification.

BRIEFING 23W

A 410-kilogram transfer batch fails one flight-specification line; rejecting it leaves only enough time to remake it if the decision is made now.

WHAT YOU DECIDE 8W

Decide whether this batch goes into the vehicle.

14.1 The line that fails

REACTOR HALL • A MODULE • PROTOCOL

SCENE 50W

The loading crew is waiting at the methane umbilical. Bhatt has four assay lines beside four flight limits. Three are inside specification; water is 400 ppm against a 50 ppm limit. The batch can be remade before Sol 475 if it is rejected now; it cannot be removed after loading.

GUIDE 68W

A specification is not one purity score. It is four separate limits, and each one is there to protect something different. So take the lines one at a time. Compare each against its own limit first. Then, for anything that is over, ask the question that actually matters. Where does that substance end up as the stream gets colder? And does that place have moving parts in it?

ASK 11W

Match each assay line to what it means for the vehicle.

BEHIND THE BUTTON 185W

Why a specification is a list and not a score. Each line guards a different piece of hardware against a different failure. There is no way to average them. Being well inside three limits earns nothing against being outside the fourth, because the fourth is the one protecting the valve. A single purity number would hide exactly the information the loading crew needs.

Why the same amount of a substance can be harmless or fatal. It depends where it ends up. Carbon dioxide freezes out in the tank, sitting still, doing nothing. Water freezes at a far higher temperature, so it turns solid in the plumbing that the valves live in. Same physics, different address. One is ballast and the other stops a valve moving.

Why an impurity can be a performance cost rather than a hazard. Carbon monoxide burns, but it releases far less energy per kilogram than methane does. So carrying it is carrying dead weight: it takes tank volume and gives little back. That is a reason to have a limit on it. It is not a reason to stop a loading.

TAKES AS READ

water freezes at a temperature the vehicle's valves reach in use • carbon monoxide does not burn to give the energy methane does • stoichiometry as a feed rate: turning a production target into kilograms in — taken as read

VERDICT 127W

A specification is not one purity score. Each line protects a different piece of hardware. Water is the failure here: 400 ppm against a limit of 50, so eight times over. And water freezes at a temperature the vehicle's valves reach in use, which puts solid ice in something that has to move. Carbon monoxide and carbon dioxide are both inside their limits. Carbon monoxide would be dead weight if it rose; carbon dioxide freezes out in the tank, where nothing moves. So each line has to be read

against its own limit and its own failure, before the valve opens. Because the water line is outside its own limit and loading is one-way, this batch is held and remade rather than averaged against three passing lines.

TAKEAWAY 16W

A batch outside one flight-critical specification line is rejected even if every other line is excellent.

14.2 How much oxygen a tonne of methane needs

COLD END • A PERSON • BALLPARK

SCENE 47W

Bhatt writes the vehicle loading numbers beside the chemistry check. Complete combustion of methane would take 4.0 kg of O₂ per kilogram of CH₄, while the flight tanks are specified at 23,000/6,600 $\approx$ 3.48 kg/kg. She wants the room to say why both numbers can be correct.

GUIDE 81W

Bhatt wants a mass, and the balanced equation gives moles. That gap is the whole question. Two moles of oxygen burn one mole of methane. But a mole of oxygen weighs 32 grams and a mole of methane only 16. So the mass ratio is not 2. Let the molar masses do the converting. Five numbers are on the board and two belong to other questions. And note in passing that a real engine runs away from this ratio on purpose.

ASK 18W

Why does the flight loading ratio ($\sim$ 3.48 kg O₂/kg CH₄) differ from the 4.0 kg/kg complete-combustion stoichiometric ratio?

BEHIND THE BUTTON 195W

Why a mole ratio is never a mass ratio. Balanced equations count molecules, because that is how reactions happen. Tanks are filled by mass, because that is what a pump moves. The two only line up if the molecules weigh the same, and they almost never do. So a 2-to-1 mole ratio here comes out as 4-to-1 by mass, and the factor of two is the difference in molar mass.

Why this is worth checking independently. The oxygen tank and the methane tank are filled from different plants on different sols. Nothing on the vehicle notices a mismatch until the engine runs, and by then the wrong answer has flown. So a second person does the arithmetic from the balanced equation rather than from the loading order, and the two are compared.

Why the engine does not use this number exactly. Complete combustion is 4-to-1 by mass, and rocket engines usually run with a little less oxygen than that. Slightly fuel-rich exhaust is cooler, which the engine's own metal appreciates, and the lighter molecules leave faster. That is a design choice about performance. It does not change the stoichiometry the loading team has to know.

TAKES AS READ

a balanced equation gives the mole ratio for a reaction • a molar mass converts a number of moles into a mass

VERDICT 81W

The balanced chemistry gives a reference: $\text{CH}_4 + 2\text{O}_2 \rightarrow \text{CO}_2 + 2\text{H}_2\text{O}$, so complete combustion is 64/16 = 4.0 kg of oxygen per kilogram of methane. The vehicle specification is about 23,000/6,600 $\approx$ 3.48 kg/kg, deliberately fuel-rich relative to that complete-combustion value. Rocket engines choose an operating mixture ratio

for performance, temperatures and hardware constraints. That design choice does not make the stoichiometric calculation wrong; it tells the loading team which number belongs to chemistry and which belongs to flight.

TAKEAWAY 15W

Stoichiometry supplies a chemistry reference; the vehicle's flight mixture ratio is a separate engineering specification.

14.3 What limits the current

ELECTROLYSIS HALL · A MODULE · CHOICE

SCENE 26W

Sundqvist wants the stacks run 20% above rated current for the sols that remain. Achebe has the measured stack-voltage curve and cooling-loop capacity beside the request.

GUIDE 57W

Faraday's law says more charge can make proportionally more product; it does not say the hardware can carry arbitrary current. As current rises, electrode and ohmic losses require more cell voltage. The thermal limit comes from the measured voltage-current behavior and the cooling loop, not from treating every volt above 1.23 V as one simple heat term.

ASK 8W

What actually limits the proposed 20% current increase?

BEHIND THE BUTTON 117W

Why Faraday's law is not the hardware limit. Charge sets a theoretical amount of product. It does not tell you what voltage the real stack needs, how hot the plates become, or whether the cooling loop can remove that heat.

Why higher current costs more than proportional electrical power. Kinetic and resistive losses push the operating voltage upward as current rises. Electrical power is current times voltage, so the input can grow faster than current alone.

Why the limit is thermal in this operating case. Achebe's measured voltage-current curve and cooling capacity say how hard this particular stack can be driven without exceeding its temperature limit. That is an engineering measurement layered on top of Faraday's chemistry.

TAKES AS READ

the stack's measured operating voltage rises with current because of kinetic and resistive losses · the cooling loop has a finite heat-removal capacity

VERDICT 84W

Faraday's law is an amount-per-charge relationship, not a current ceiling. Pushing more current can make more product, but real electrode kinetics and resistance also push the required operating voltage upward. Because electrical power is current times voltage, the stack's electrical and thermal load rises faster than current alone. Achebe therefore uses the measured voltage-current curve and cooling-loop capacity to set the ceiling. The chemistry tells how much product a coulomb can buy; the hardware tells how quickly the station can safely spend those coulombs.

TAKEAWAY 23W

Faraday's law sets product per coulomb; the stack's voltage-current curve and cooling system set how many coulombs per second the hardware can survive.

Irreversible loading decisions are made against each specification line, with enough time left to recover from a rejection.

Release the full tank

PLAN CARD 89W

Sol 475. Eleven sols before the return window, the ascent vehicle has 6.6 tonnes of methane and 23 tonnes of oxygen aboard. The rejected Sol 445 batch never entered the vehicle, every loaded batch has an assay, the water-breakthrough alarm is armed, and the final catalyst charge has held through the qualified clean-feed operating window. Today you freeze the rule the next crew inherits, state exactly what the catalyst evidence supports, and make the one-way decision this whole season has been building toward: release the propellant set for flight.

BRIEFING 25W

The target masses and final assays are in; the last job is to decide whether the propellant set is ready to become a flight system.

WHAT YOU DECIDE 14W

Say what is known, how well it is known, and what was never established.

15.1 Write the rule before the trace moves

WATER PLANT · A MODULE · TRIGGER

SCENE 38W

Abiola is freezing the water rule before flight release. He loads the historic conductivity breakthrough into the simulator. Once the replay starts, the thresholds are locked; the same rule will remain for the crew that inherits the plant.

GUIDE 84W

One rule, written before the trace moves, and fixed the moment you release it. Two numbers set it. The changeover takes one update to complete once it is called. And past $3 \times$ baseline the feed water is outside what the reactor will accept. So work backwards from 3 by however far this trace climbs in a single update. Then check the other direction too. Set the line too low and a perfectly good bed gets swapped out for a reading that meant nothing.

ASK 20W

At what conductivity do you move the lag column into lead service, given the changeover takes one update to complete?

BEHIND THE BUTTON 209W

Why conductivity is the signal to write the rule on. It climbs as the column runs out of capacity and starts letting ions through. So it moves *before* the water is out of specification, not after. That gap is the only reason a rule is possible at all. A measurement that only told you after the damage would be a record, not a warning.

Why a threshold has to allow for lead time. The swap is not instant. Call it and one update passes before the fresh bed is in service, and the trace keeps climbing through that update. So the line cannot sit at the limit. It has to sit far enough below the limit that the climb during the swap still lands underneath it. Work backwards from the consequence, never forwards from the reading.

What writing for a crew you will never meet changes. Nobody will be there to exercise judgement when it fires. So the rule has to hold for a trace nobody has seen, including one that rises faster than this one. That argues for margin. But margin costs columns, and swapping a clean bed wastes the very capacity the rule exists to protect. Both errors are real, which is what makes this a decision.

TAKES AS READ

an exchange bed reaches breakthrough without warning · a lead and lag pair lets one bed be changed while the other works

VERDICT 104W

The outlet stays near baseline while the front is still inside the bed, then climbs fast once it reaches the end. So a threshold is only useful if the response can finish before the water leaves specification. Here the swap takes one update. Fire at about twice baseline and it completes while the next reading is still 2.6. Wait for 2.6 and it completes after the trace has already crossed 3.0. That is the whole calculation: work

back from the limit by one update's worth of climb. And write it before the replay, so the next crew inherits a rule rather than somebody's hindsight.

TAKEAWAY 23W

A breakthrough warning is only useful if its trigger is written before the trace rises and leaves enough lead time for the swap.

15.2 The honest claim about the bed

CATALYST BAY · A MODULE · CHOICE

SCENE 34W

Kaur is freezing the catalyst record on Sol 475. The last spare charge has run 175 sols since commissioning, inside the stated temperature/flow envelope, while the polishing outlet stayed below its halide warning line.

GUIDE 65W

Kaur has a long run and four claims. Ask what conditions were actually varied. The charge saw the qualified temperature and flow envelope and 175 sols of clean feed. It did not see the chlorine-bearing contamination that killed the previous charge. A strong record supports a stronger duration claim than five sols did, but still cannot prove resistance to a poison that was deliberately excluded.

ASK 10W

What is the strongest claim the 175-sol record actually supports?

BEHIND THE BUTTON 98W

Why duration strengthens one claim and not every claim. One hundred seventy-five sols of stable conversion is strong evidence about durability inside the tested clean-feed operating envelope.

Why absence of a poison still cannot prove resistance to it. The water train was repaired precisely so halide breakthrough would not reach the reactor feed. The new run therefore contains no controlled test of chlorine-poison tolerance.

Why conditions stay attached to the handover. Temperature, flow, pressure and feed quality define the operating envelope in which the result was observed. Removing them turns a supported statement into a broader unsupported one.

TAKES AS READ

a claim is limited by the conditions it was tested under · a poison accumulates over time rather than acting at once · rate constants, and how steeply they climb with temperature — taken as read

VERDICT 62W

The last charge has held stable conversion for 175 sols inside the qualified temperature, pressure, flow and clean-feed envelope. That is real durability evidence and should be stated plainly. It still does not show resistance to chlorine-bearing contamination, because the repaired water train kept that challenge out of the feed. The strongest defensible handover therefore includes both the duration and the conditions.

TAKEAWAY 21W

A long run strengthens the claim inside the tested operating envelope; it does not create evidence for conditions that were excluded.

15.3 Release the full tank

PLANT CONTROL · A PERSON · CHOICE

SCENE 54W

Wells puts the final board beside the ascent vehicle telemetry. Methane: 6.6 tonnes. Oxygen: 23 tonnes. Every loaded batch is inside its own flight limits; the Sol 445 batch was rejected and remade. The conductivity trigger is armed, and the catalyst record is qualified rather than universal. The return window opens in eleven sols.

GUIDE 58W

Do not ask whether every uncertainty on Mars has disappeared. Ask whether the flight decision's required claims are supported. The tank masses are measured. Batch acceptance is documented. The rejected batch stayed out. The water-control rule is armed. The remaining unknowns concern future column life, deeper ice and untested catalyst-poison conditions—not the specification of the propellant already aboard.

ASK 10W

What do you sign eleven sols before the return window?

BEHIND THE BUTTON 104W

What release means. This is not a claim that the plant is solved forever. It is a decision that the propellant set now aboard satisfies the vehicle's stated mass and chemistry requirements.

Why unresolved science can remain. The next crew still needs better breakthrough statistics and deeper-ice chemistry. Those uncertainties belong to future production. They do not automatically invalidate measured, in-specification propellant already loaded.

Why the rejected batch matters. Readiness includes the decisions not taken. The Sol 445 water-contaminated batch never entered the vehicle, and the replacement production is included in the final mass. The audit trail therefore closes mass, chemistry and reversibility together.

TAKES AS READ

the vehicle flight specification is the acceptance rule for propellant already loaded · all loaded batches have traceable assays and the displayed tank masses are reconciled

VERDICT 104W

The required flight claims are now supported independently. The tank masses are reconciled at 6.6 tonnes methane and 23 tonnes oxygen. Every batch that entered the vehicle passed its own specification; the batch that failed did not enter. The water system has a prewritten breakthrough response, and the catalyst record states only the conditions it actually survived. Waiting for every unknown about deeper regolith or future column lifetime would not improve the chemistry of propellant already measured aboard. Releasing without conditions would overstate what the season proved. The defensible action is to release this propellant set to flight and carry the remaining limits forward.

TAKEAWAY 21W

Readiness is a bounded claim: these measured tanks meet these flight requirements, with the remaining unknowns left attached to future production.

END OF THE SOL 22W

Readiness means the required mass is aboard, every loaded batch meets its limits, and the remaining uncertainty is stated rather than hidden.

THE CLOSING CARD

The fill finished on sol 475, eleven sols before the window opened, with six point six tonnes of methane and twenty-three of oxygen aboard and every batch signed against the assay it came with. What had been holding the plant down was never the set point both sides spent a fortnight arguing about: the polishing column broke through around sol 261 and let halide contamination into the water/electrolysis train for thirty-eight sols; chlorine-bearing carryover reached the reactor feed, and the catalyst died from the inlet end while the temperature profiles were being read as heat damage. The last spare charge went in on sol 300 and held 71% conversion to the end.

What it cost: eleven sols of production, one catalyst charge with no replacement on the planet, and about four hundred kilograms of oxygen out of the vent while a radiator stood under a fortnight of storm dust. What is unfinished: the lead-and-lag columns are plumbed and the interval between changes is a guess taken from one breakthrough; the conductivity alarm was wired to the plant panel before flight release and has never sounded; and nobody has measured what this ground does to a catalyst at the deeper cut, which is the ice the next crew will be working.

Six people fly home because of you. You found a poisoned catalyst that everybody else was reading as heat damage, you spent the last charge on the planet at the right sol rather than the frightening one, and you signed every kilogram against an assay you had checked yourself. The rocket lifts full. You are the one who filled it.

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